Portable Single-Mirror Schlieren Apparatus

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Design Description

Revision: Nov 7, 2026

This document describes the design of the apparatus, and is maintained authoritatively as Markdown fragments in the project repo on GitHub at https://github.com/fritzm/schlieren/tree/main/docs/design. It serves as durable context for AI assistants to reason about the design, as well as being a human-readable guide.

Conventions

- The design described here is the baseline. Unless marked otherwise, dimensions, parts, and methods are settled.
- **Provisional** marks a design area, dimension, or method that is still open or evolving, typically pending a physical fit check, measurement, or further design. Provisional areas are rolled up with next steps in §11.
- Measured values, manufacturer specifications, and calculated values are identified as such where the distinction matters; a measured value takes precedence over a catalog value.
- Quantities, vendors, part numbers beyond those needed to define the design, procurement state, and the CAD state of printed parts are recorded in the associated BOM at (`bom/bom.csv`), not here.

1. Project context and design constraints

The apparatus is a portable, single-mirror schlieren system intended initially for several days of vacation/travel use in a location with a large room and a substantial tabletop. It should remain useful afterward as a bench-top laboratory instrument.

The complete system should pack into a standard airline rolling suitcase. Plan around the ordinary 50 lb / 23 kg checked-bag target.

The design is cost-conscious rather than constrained to a fixed dollar ceiling. Commodity 2020 extrusion, Baltic-birch plywood, standard fasteners, and printed PLA are preferred where they are adequate. Commercial optomechanics are used where they materially simplify alignment or provide precision that would otherwise be troublesome.

1.1 Fabrication envelope

Near-term 3D printing is PLA only. Critical optical heights and precision datums should be established by metal posts, optical hardware, shim stock, mechanical stops, or adjustment mechanisms rather than relying solely on as-printed PLA dimensions.

The following are out of scope:

- custom machined parts requiring a mill or lathe;
- cutting stainless-steel stock or sheet to custom shape or length.

Use off-the-shelf metal parts, vendor-supplied finished dimensions, plywood, thin foil, or printed PLA. Drilling and tapping of aluminum parts is within scope.

2. System overview and physical layout

The apparatus consists of two main physical assemblies:

1. a tabletop optical head; and
2. a separate mirror cell, normally tripod-mounted approximately 3.2 m away.

The tabletop optical head uses two independently yaw-adjustable 2020 aluminum-extrusion rails mounted to a common Baltic-birch front pivot plate. The rails diverge slightly away from the mirror-facing end so that each rail follows one optical chief ray:

- the source rail carries the illumination system directed toward the mirror;
- the imaging rail carries the cutoff and camera system for the return beam.

The source rail contains, in optical order:

1. light source (with interchangeable threaded LED modules, condenser lens, and adjustable iris)
2. adjustable slit fixture
3. field lens, on a post of its own just ahead of the slit (§7.9)

The slit is the effective optical source, with the LED illuminating it through the condenser and iris. The imaging rail contains, in optical order:

1. cutoff carriage at the returned slit-image plane;
2. telephoto lens in adjustable cradle;
3. iPhone attached to lens, with additional rail support.

A spherical mirror in a separate tripod-mountable cell forms an image of the source slit back near the optical head. Refractive-index gradients in the test region near the mirror deflect portions of this return beam relative to the cutoff, producing schlieren contrast.

3. System-level geometry

3.1 Primary optic

- Mirror: Skyoptikst D203F1600
- Diameter: 203 mm
- Focal length: 1600 mm
- Surface: spherical
- Radius of curvature / nominal operating distance: approximately 3200 mm

The source slit and imaging/cutoff system operate near the mirror center of curvature. The test region is near the mirror, where outgoing and returning beams remain close together.

The returned slit image defines the cutoff plane.

3.2 Source cone

At 3200 mm, the mirror subtends a half-angle of approximately:

$$\theta = \tan^{-1}(101.5/3200) \approx 1.82^\circ$$

The intended illumination cone is approximately ±1.82° (the mirror's half-angle, just filling it) to ±2.0°, corresponding to approximately NA = 0.032–0.035 in air. Slight mirror overfill is preferred to underfill.

Some beam diameters of the nominal ±1.82° cone for reference:

Distance from source slit	Approx. beam diameter
1 m	63 mm
2 m	127 mm
3.2 m	203 mm

The mirror's approximately f/7.9 focal ratio should not be confused with this source cone. The source operates near 2f, at approximately the mirror center of curvature.

3.3 Source/imager offset

The two rail pivot centers are separated by 95 mm at the mirror-facing front plate. At 3.2 m this gives approximately 1.70° total chief-ray separation, or rail yaw of 0.85° from the mirror normal.

Each rail points directly toward the mirror and follows its own chief ray. Sliding a fixture along a rail is therefore predominantly an axial adjustment.

The source slit and cutoff need not occupy exactly the same axial position. Near 2f,

$$\frac{ds_i}{ds_o} \approx -1$$

so small source-slit displacement produces approximately equal and opposite conjugate-image displacement. This can be an aid in avoiding mechanical interference between fixtures on opposite rails.

A useful starting point is about 30–40 mm total slit/cutoff axial staggering, obtained by placing them approximately ±15–20 mm around nominal 2f. This is an experimental setup adjustment rather than a fabrication-critical dimension.

3.4 Optical axis

The common optical axis is 72.35 mm above the top surface of the 2020 extrusion, established by commercial optomechanics to simplify alignment. The stack-up is:

0.254 mm	steel datum shim on rail (bridges the rail top channel)
50.000 mm	Thorlabs TR50/M post
22.098 mm	various Thorlabs SM1 metric fixed mounts
72.352 mm	

3.5 Coordinate frames

For optical-head CAD:

- take the rail-top surface as z = 0;
- do not reference optical height to the tabletop or support fixtures.

The optical head uses three coordinate frames with common senses:

- +y forward, toward the mirror;
- +x to the right when looking down from above with the mirror ahead;
- +z up, with the rail top at z = 0.

These frames are right-handed and differ only in where their y axes lie:

- pivot frame: y along the centerline of the front pivot plate, with the origin at the midpoint of its aft edge;
- source rail frame: y along the source rail centerline;
- imaging rail frame: y along the imaging rail centerline.

Each rail frame yaws with its rail. Its origin is on the rail top at the rail's pivot axis; a fixture is described with its own rail station at y = 0. In a rail frame the optical axis is at x = 0, z = 72.35 mm, and pitch, yaw, and roll are rotations about x, z, and y respectively. A rotating fixture may define a local frame, stated where it is used.

4. Tabletop optical frame

The tabletop frame provides a portable, adjustable base to support the optical head fixtures. It is a weight- and budget-conscious alternative to an optical bench or breadboard. The model is `src/schlieren/parts/frame.py`.

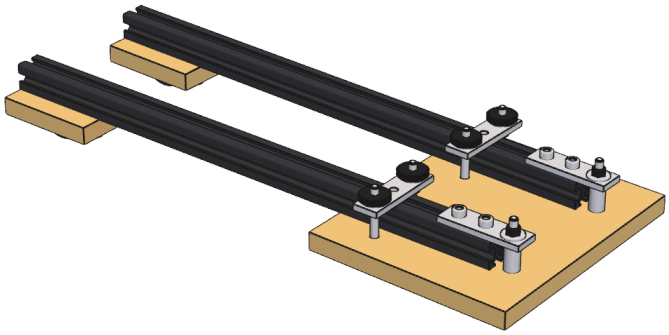


Fig. 1. Tabletop frame: rails, front pivot plate with pivots and yaw clamps, and rear support blocks

4.1 Rails

- IXGNIJ black-anodized metric 2020 extrusion, Amazon B08Y8N7FD1
- 20 × 20 × 400 mm
- two rails used
- nominal 6 mm European-style slot

Ordinary attachments use M5 × 0.8 socket-head hardware with flat washers in standard metric lengths and roll-in/drop-in M5 T-nuts (Amazon B0DP6JBX4Y).

Sighting toward the mirror, the source rail is on the left (negative x in the pivot frame) and the imaging rail is on the right. This is dictated by fact that the iPhone extends well to the right of its camera axis when sighting along that axis toward the mirror (§9.1); placing that overhang toward the outside of the optical head keeps the phone clear of the source rail and its fixtures.

4.2 Front pivot plate

The front pivot plate is a 180 × 150 mm Baltic-birch rectangle, with its long dimension is transverse to the rails.

Plate coordinates are the pivot frame (§3.5): the origin is at the midpoint of the aft edge, +y runs toward the mirror along the plate centerline, and +x runs to the right seen from above with the mirror ahead. The mirror-facing edge is at y = 150 mm, and the left rail is the one at negative x.

The pivot line is 25 mm behind the mirror-facing edge. With a pivot separation of 95 mm, this gives pivot centers at:

- left pivot: (-47.50, 125.00) mm;
- right pivot: (+47.50, 125.00) mm;

Each yaw-clamp station is 90 mm aft of its pivot along its rail axis. At the nominal 0.85° splay, this gives yaw-strap centers at approximately:

- left strap center: (-48.84, 35.01) mm;
- right strap center: (+48.84, 35.01) mm.

Each of the fixed yaw clamp straps is perpendicular to its rail. At the nominal 0.85° splay, this gives clamp bolt locations at approximately:

- left outer: (-68.83, 35.31) mm;
- left inner: (-28.84, 34.71) mm;
- right inner: (+28.84, 34.71) mm;
- right outer: (+68.83, 35.31) mm.

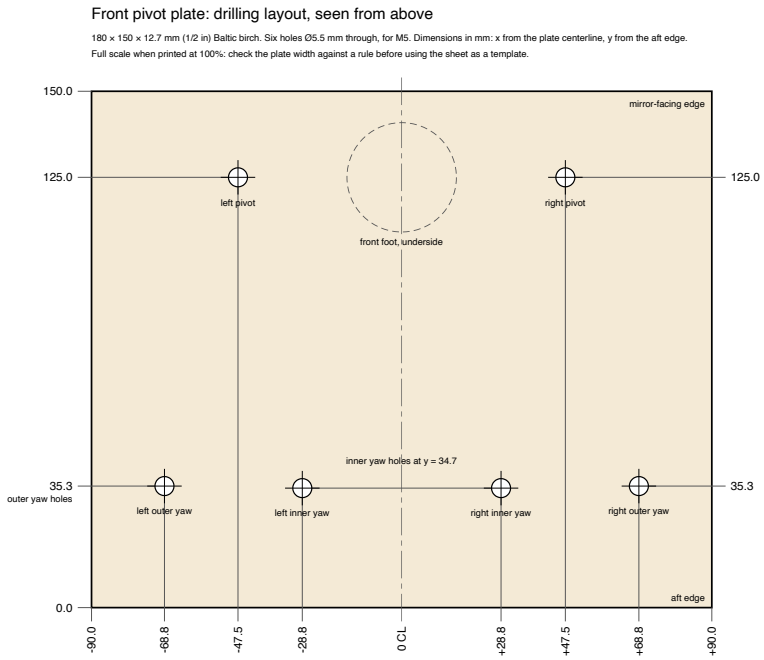


Fig. 2. Drilling template for the front pivot plate, in plate coordinates

With 40 mm bolt spacing and 90 mm pivot-to-clamp distance, the achievable yaws adjustment range of each rail is slightly more than ±3°.

A front Sorbothane support foot is attached below the plate, centered at approximately (0, 125) mm, directly between the two rail pivots. This keeps the support point on the pivot line and is clear of the pivot and yaw hardware.

1/2 in Baltic-birch plywood sheet stock is shared across the tabletop frame and mirror cell: it supplies the front pivot plate, the two rear rail-support blocks, the mirror-cell stationary plate, and the mirror-cell base plate.

4.3 Pivot and yaw hardware

Pivot and yaw hardware use commercially available 2020 three-hole joining plates:

- MOPFOL 2020 3-hole joining plate, Amazon B0G8DCS7VP
- approximately 60 × 18 × 4 mm
- 20 mm hole pitch
- four in all: two as pivot lugs and two as yaw clamp straps.

Pivot stack

Per rail, from the bottom up:

socket head → oversized washer → plywood pivot plate → 20 mm spacer → 4 mm joining plate → oversized washer → M5 nyloc

The pivot bolts pass through 5.5 mm clearance holes in the plywood. The 20 mm spacer holds the pivot lug at the rail top, where the lug lies along the rail with the pivot bolt through its front hole; its other two holes fix it to the rail by M5 × 8 socket-head screws from the general M5 assortment into T-nuts in the rail top slot.

Hardware:

- McMaster 92290A265, M5 × 50 partially threaded 316 SS socket-head screw
- McMaster 92871A091, 20 mm × 10 mm OD × 5.3 mm ID stainless spacer
- McMaster 91116A350, M5 oversized washer, 15 mm OD
- McMaster 93625A225, M5 nylon-insert locknut

Provisional: pivot bushings

Identified during construction: the nyloc is left loose enough for the rail to turn, so nothing preloads the pivot stack; with 5.5 mm clearance holes in the plywood and lugs the bolt has about 0.25 mm of radial play at each end and can tilt. The yaw clamps hold the rail against sliding across the plywood effectively, but they resist turning the rail about the clamp center only weakly, and the long rear rail multiplies any lateral play at the front pivot (play × distance aft of the clamp / 90 mm). The end result is some undesireable position wobble in the rails even when firmly clamped.

A proposed remediation is to bore the plywood and lug clearance holes out to 8 mm and press fit plain (flangeless) sleeve bushings. These would locate the bolts laterally at both ends of the pivot to about 0.02 mm and hold the bolts square to the plywood and lugs. The bushings carry no axial load, and rotation within them is minimal occurring only while the rail yaw is being actively adjusted.

Hardware:

- McMaster 6658K724, 5 × 8 × 10 mm oil-embedded bronze sleeve bushing
- McMaster 6658K116, 5 × 8 × 4 mm oil-embedded bronze sleeve bushing

Yaw clamps

One 3-hole joining plate spans transversely across each rail as its yaw clamp strap. Its two outer holes take the yaw bolts, one on each side of the rail; the rail rotates beneath the strap.

A strip of McMaster 9852N37 adhesive-backed 1/32 in EPDM is applied to underside of each strap and provides friction where the clamp strap bears against the rail top. The strip spans the strap width and stays between the outer bolt holes (about 30 mm long).

Each yaw bolt stack, from the bottom up:

socket head → oversized washer → plywood pivot plate → (alongside the rail) → EPDM friction strip → 4 mm yaw strap → oversized washer → M5 thumb nut

The yaw bolts pass through 5.5 mm clearance holes in the plywood. Yaw is adjusted and locked at the front pivot plate by loosening the two thumb nuts, rotating the rail beneath the strap, and retightening.

Hardware:

- McMaster 92290A265, M5 × 50 partially threaded 316 SS socket-head screws — same part as the pivot bolts
- McMaster 91116A350, oversized M5 washers
- McMaster 92815A202, M5 low-profile knurled thumb nuts
- McMaster 9852N37, 1/32 in EPDM friction strips, cut from same stock as used elsewhere in the project

4.4 Rear supports and feet

Each rail has a removable 75 × 50 mm Baltic-birch rear rail-support block. The 75 mm dimension runs along the rail and the 20 mm rail is centered across the 50 mm block width.

Each block attaches to the rail bottom slot with two M5 socket-head screws into roll-in/drop-in M5 T-nuts. The two screw holes lie on the rail centerline and are 50 mm apart, symmetrically placed about the block center; with the 75 mm block length, each screw center is 12.5 mm from its nearest end.

Each block's screw stack, from the bottom up:

socket head → two flat washers → plywood rail-support block → M5 T-nut in the rail bottom slot

The doubled washers provide some margin for a standard 20 mm screw to avoid bottoming out against the rail slot floor in the face of minor plywood thickness and rail manufacturing variations.

A Sorbothane foot is centered on the underside of each block, directly beneath the rail centerline and midway between the two attachment screws. The complete frame is therefore supported on the table/bench top at three points:

- one centered foot beneath the front pivot plate;
- one foot beneath each rear rail-support block.

Moderate static compression of the feet is acceptable; the resulting small pitch and roll effects can be corrected with fine mirror alignment. The selected feet are:

- McMaster 8215K2, adhesive-back sorbothane bumper, 1-1/4" OD, 4 lbs. max load
- McMAster 8215K6, adhesive-back sorbothane bumper, 1-1/4" OD, 10 lbs. max load (alternate, stiffer)

5. Rail-mounted posts

A common post design establishes fixtures accurately at the optical axis, and bridges between the commercial optomechanical and 2020 rail ecosystems. The model is `src/schlieren/parts/rail_shoe.py`.

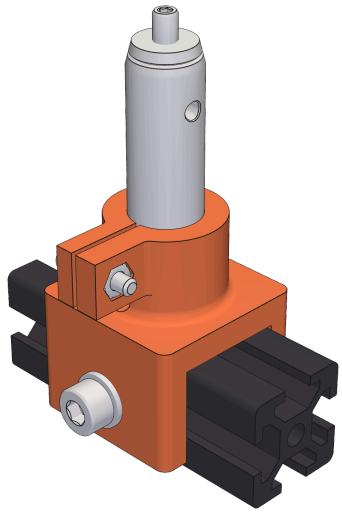


Fig. 3. Common rail-mounted post design

5.1 Post and shim

The chosen optical post is the Thorlabs TR50/M. These are 303 stainless steel, 12.7 mm diameter, 50 mm long with an M4 setscrew at the top which accommodates various Thorlabs SM1 metric fixed mounts.

Each post rests on a stainless steel shim disk, which provides a consistent metal seating surface across the central slot in the 2020 rail. The shims are McMaster 2895T62, 3/4 in OD discs × 0.010 in ±0.0005 in thick, full-hard 18-8 stainless steel.

5.2 Rail shoe

The shoe is a printed PLA U-shaped saddle that:

- straddles the 2020 extrusion;
- locks with M5 hardware into T-nuts in the rail side channels;
- leaves the rail top available as the datum surface;
- carries a vertical 12.7 mm through-bore for the TR50/M post.

The post passes through the printed shoe and bottoms on a stainless datum shim resting on the rail top, so the manufactured steel parts, and not the printed PLA, establish the precision optical height.

The post bore incorporates a single-split clamping collar so that the post can be locked against rotational movement while remaining fully seated on the metal datum. Ear thickness and the approximately 1–2 mm relaxed split gap are designed around the clamping setup:

M3 socket-head screw → clearance through one ear → captured M3 hex nut in the opposite ear

The shoe's ears point outboard, away from the other rail, so the clamp is reached from the side of the apparatus: toward the right on the imaging rail, and (the shoe turned half a turn about its post) toward the left on the source rail. The SM1RC/M split rings on the source rail's posts (the slit head's and the field lens's) are turned the same

way, with their locking screw heads outboard. The cutoff carriage's ring and the lens cradle's collar clamps face outboard on the imaging rail.

Hardware:

- McMaster 91292A114, M3 × 12 fully threaded 18-8 SS socket-head screw
- McMaster 91828A211, M3 18-8 SS hex nut

5.3 Post-top mounts

Four fixtures use the common post/shoe architecture:

1. light source (§6);
2. source-slit fixture (§7);
3. field lens tube (§7.9);
4. cutoff carriage (§8).

The light source uses a Thorlabs SMR1/M internally threaded lens mount, and the source-slit, field-lens, and cutoff assemblies seat in Thorlabs SM1RC/M lens tube split slip rings. These mounts all preserve a 22.1 mm post-top to optical-axis dimension.

6. Light source

The light source is a single assembly utilizing the common post support (§5). An internally threaded mount carries an interchangeable LED module on its rear face, and a condensing lens and iris on its forward face. The condenser images the LED emitter onto the source slit (§7) ahead, and the iris sets the angular size of the illumination cone. The model is `src/schlieren/parts/light_source.py`.

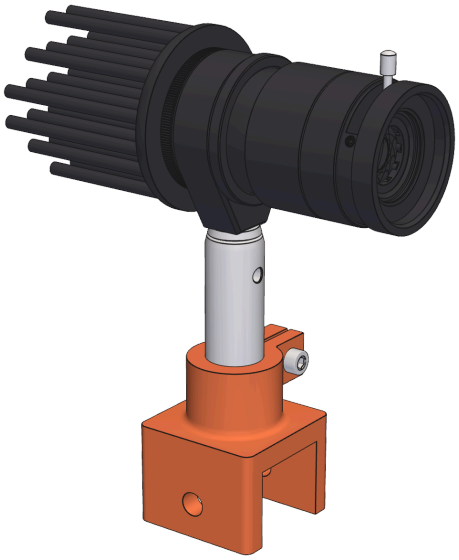


Fig. 4. Light source on its post and rail shoe

6.1 Assembly and stack order

From the LED toward the slit:

1. LED module;
2. Thorlabs SMR1/M mount;
3. Thorlabs SM1V05 adjustable lens tube;
4. Thorlabs ACL2520U-A condensing lens seated in the lens tube;
5. Thorlabs SM1L03 spacer ring, to provide relief between the lens vertex and the iris ahead;
6. Thorlabs SM1D12 adjustable iris.

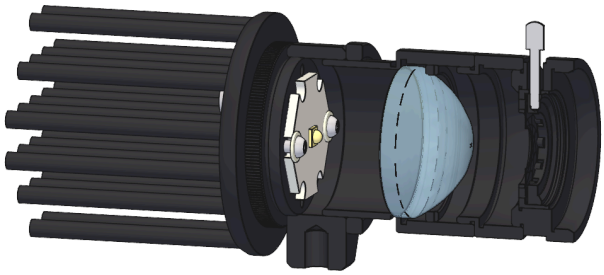


Fig. 5. Light-source cutaway, showing internal position of LED carrier and condensing lens

The LED modules and the condenser lens holder share opposite sides of the SMR1/M mount thread. The module threads in until seated against the rear SMR1/M face, providing reapeatable LED positioning, and the remaining forward thread is used by the lens holder for focus adjustment.

6.2 LED modules

The LED modules are constructed around Thorlabs SM1CP2M SM1 externally threaded solid aluminum end caps. Each cap is drilled and tapped so an LED carrier board can be screwed to its front face and a heatsink screwed to its rear face. The cap's external SM1 thread then serves as the module interchange standard, supporting and aligning the module in the assembled light source.

Each LED module has a pair of 18 AWG power leads exiting between its heatsink pins columns, terminated in a Molex Micro-Fit 3.0 connector. A mating pigtail connects via banana-plugs to an ALIENTEK DP100 30V 5A adjustable CC/CV DC power supply (Amazon B0CWRG6YFM).

LEDs

The LEDs chosen are on industry-standard 20 mm "star" MCPCB carriers:

Module	Emitter	Board	Initial operating current (mA)
Green	OSRAM OSLON SSL 120, 530 nm nominal	NewEnergy star board, LST1-01F06-GRN1-00	~350
White	Nichia 519A R9080, 5000 K	Convoy 20 mm star/DTP carrier	~500–700

Each star board is fastened to its module with a pair of M2 x 6 mm screws through board slots into tapped holes on the front of the cap; slightly oversized board slots permit final centering of the emitter on the cap axis. A thin insulating washer goes under each screw head, keeping the metal head off exposed board contacts and spreading clamp contact over the board slot. A thin layer of thermal compound is used between board and cap.

Power leads drop past the star board's flat edges and pass out through two Ø2.0 mm holes in the cap and heatsink base, positioned inside the cap thread relief.

The white LED dissipates about 2 W at 700 mA, for a predicted steady state temperature rise roughly 14 °C above ambient. The green module at 350 mA dissipates about 1.1 W. **Provisional:** bench-test the predicted temperature rises.

Heatsink

- Alpha CN40-40B natural-convection pin-fin heatsink, one per module
- 6063 aluminum, black anodized; 24.6 g
- Ø40.0 mm × 40 mm overall, 3.0 mm flat base, Ø2.8 mm pins on a 6.9 mm square grid
- manufacturer natural-convection rating 6.9 °C/W

The above rating is the manufacturer's pins-up test; the module runs with pins horizontal, which pin fins tolerate well.

The heatsink and cap are fastened with three M3 × 6 mm socket-head screws run through drilled clearance holes in the heatsink base into corresponding tapped M3 holes in the cap. The screws are positioned so their heads seat against the rear of the heatsink base in open cells between sets of four pins. Three such screws hold the heatsink without a rocking axis. A thin layer of thermal compound is also used between the heatsink and the cap.

The heatsink base also has drilled holes to accommodate exit of the LED power leads, which are run out between pin columns.

Cap and heatsink hole layout

The drill and tap prescriptions for the cap and heatsink are as follows. Coordinates are in millimeters from the optical axis, +x to the right and +y up as seen from the LED side, in the frame of the heatsink pin lattice:

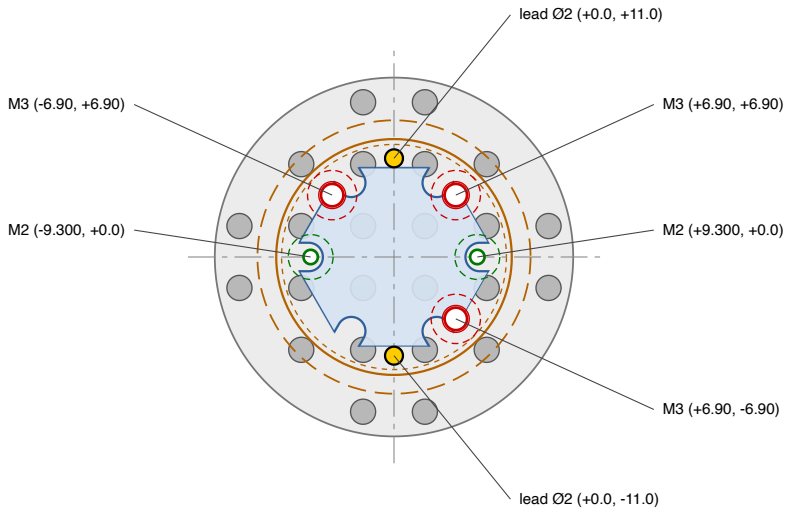


Fig. 6. Cap and heatsink hole layout, seen from the LED side. Grey: heatsink; orange: cap; blue: 20 mm star; red: M3 hardware; green: M2 hardware

Hole	Position	In the cap	In the heatsink base
M3, three	(+6.90, +6.90), (−6.90, +6.90), (+6.90, −6.90)	Ø2.5 tap drill, M3 × 0.5	Ø3.3 clearance
M2, two	(±9.30, 0)	Ø1.6 tap drill, M2 × 0.4	none
Lead, two	(0, ±11.0)	Ø2.0	Ø2.0

All holes in the cap and in the heatsink base are **through holes**, drilled from one face. Plain plug taps cut full threads through them; no bottoming taps are required. No screw reaches the far face of the cap in either direction: the M2 screws are 6 mm under the head, about 4 mm of it in the cap from the front past the board and washer, and the M3 screws engage about 3 mm into the cap from the back past the 3.0 mm heatsink base.

Drilling fixtures

In practice, the cap and heatsink require fairly precise drilling; the lead holes' outer edges, for example, leave only about 0.55 mm to the cap's outer thread relief. The Ø2.0 mm lead hole alignment between cap and heatsink must also be kept pretty tight to avoid impinging the Ø1.7 mm leads passing through both. To this end, a bushed drill template and custom alignment nests for both the caps and heatsink were designed. The model for these is `src/schlieren/parts/led_drill_fixture.py`.

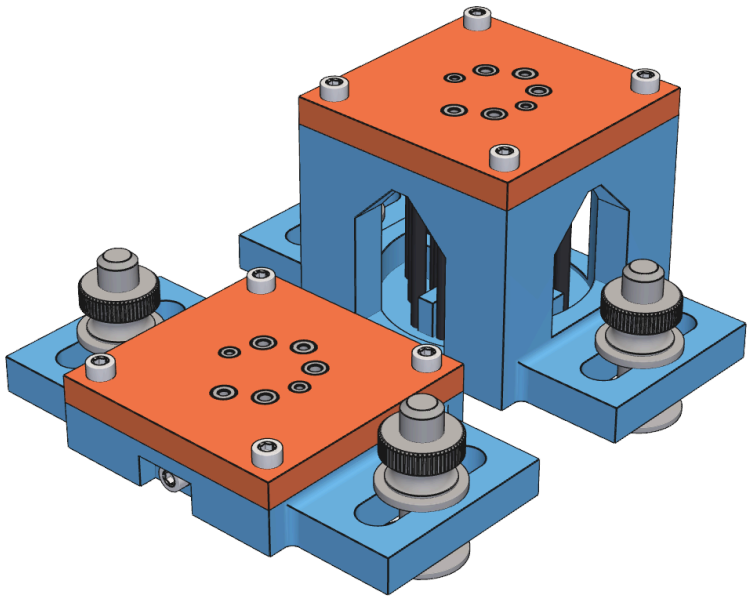


Fig. 7. Drilling fixtures: bushing plate on the cap nest (left) and on the heatsink nest (right)

A single **bushing plate** serves both parts, so the lead exit holes and the M3 holes that fix the cap-to-heatsink rotation are drilled from the same coordinates. A matching **cap nest** and a **heatsink nest** each locate one type of part beneath. All are printed in PLA, with a keyed registration between cap and nests allowing only a single assembled orientation for each.

Selected hardened steel metric press-fit drill bushings for the plate are as follows:

Hole	Count	Bore (mm)	Outside diameter (mm)	SKU
M2 tap drill	2	1.6	4	McMaster 96990A132
M3 tap drill	3	2.5	5	McMaster 96990A147
Lead hole	2	2.0	5	McMaster 96990A139

The bushing plate bores are drawn over the bushing outside diameter to allow for printed hole undercutting. They give a press fit, found with the bore coupon (`led_drill_bore_coupon`, 0.00 to 0.20 mm over in 0.05 mm steps). A little low-viscosity cyanoacrylate, wicked in around each bushing, retains it.

The bushing plate seats on each nest's top face, and is held there during drilling by four M3 × 12 mm screws through corner clearance holes in the bushing plate into brass flanged heat-set inserts in each nest's corners. The nests are designed to leave a 2.0 mm gap between the bottom of the plate and the top of the drilled surface to accommodate drill chips. A boss under the center of the bushing plate carries a disc of the 1/32 in adhesive-backed EPDM sheet (McMaster 9852N37; the same sheet used elsewhere). Seated, that disc presses down on the drilled part's top face with about 0.3 mm of compression to help restrain the part against wobble or bit grabbing.

The cap nest additionally has an M3 x 12 mm side clamp screw through a heat-set insert which bears against the knurled edge of the held cap, while the heatsink nest has a cross-shaped rib that engages between the held heatsink's pins opposite the drilling surface. Both of these measures restrain the held parts against rotation within their nests while drilling.

The nests as modeled mount on the base of a Dremel 220 drill press, which has four diagonal keyhole slots. Each nest inself has two slotted ears, on its ±x faces. The nests sit turned 45° on the drill press base so their x axis lies along one pair of base slots. One carriage bolt on each side extends upward through a drill press base slot, through a nest ear and a flat washer, and engages a flanged knurled thumb nut at the top of the stack. The fixture can thus be located in x along the base slots, and in y along the ear slots, enough to bring any bushing under the drill drill spindle.

Both types of parts are drilled through the plate from the LED-side face. To move between holes, loosen the thumb nuts, lower the bit into the next bushing and hold it there, and tighten the thumb nuts; the bushing sets the fixture's position.

- Caps: all seven holes, through. Then tap the M3 and M2 holes by hand.
- Heatsinks: the three M3 positions the two lead holes. Then open the M3 holes to Ø3.3 clearance with a hand drill at low speed.

The plate carries all seven bushings for both parts, but the M2 positions are not drilled in the heatsink.

6.3 Condenser optics

The condenser images the LED emitter onto the source slit (about 6.9× magnification). Each slit point receives light from its conjugate emitter point through the full iris aperture, so its cone is uniformly filled in angle. Emitter structure appears as intensity variation along the slit, not as structure in the mirror illumination. The LED image should cover the illuminated slit length; the domes enlarge the apparent emitters by about 1.4–1.5×. The image's brightest portion is about 10.4 mm long, as calculated, with a weaker shoulder that extends to about ±7.8 mm. The shoulder was measured at ±7.5 mm from the wall image (iris open, slit removed) at 6.6×, 153 mm from the lens vertex to the slit, and the model scales it to the present magnification (`LightSourceParameters.lit_half_length`). The clearances of the slit head and field lens (§7.2, §7.9) use that extent.

Field lens. A slit point's cone is centered on the chief ray from the center of the iris to that point, not on the mirror. The iris leaves are about 150 mm behind the slit (the 159 mm from the lens's convex vertex less the 9.4 mm to the leaves, below) and the mirror 3200 mm ahead of it, so for a point a distance h along the slit the chief ray is tilted by about h/150, and the cone's footprint on the mirror is displaced by about h (1 + 3200/150) ≈ 22 h: 116 mm at the end of the brightest portion of the slit (h = 5.2 mm) and 176 mm at the edge of the shoulder (h = 7.8 mm), more than the mirror's 101.5 mm radius. Each slit point lights a mirror-sized disk, but the disks' centers are spread along the slit, so without a correction the illumination falls along the slit to about 55 % of its central value at the mirror's edge. A field lens just ahead of the slit removes the spread (§7.9): it images the iris onto the mirror, which aims every chief ray at the mirror's center, so the cones from all slit points overlap there.

The lens is the Thorlabs LA1433-A (f = 149.5 mm), and the slit's distance from the condenser is chosen for it. With the lens at the position of §7.9, its focal length images onto the mirror an iris about 157 mm behind its object-side principal plane, which puts the iris leaves about 150 mm behind the slit. The condenser's conjugates follow from that distance (below), and in turn the gap that focuses the emitter on the slit. Every cone then centers on the mirror (paraxially, within 0.1 mm, from any point of the lit slit). A lens ahead of the slit leaves the condenser's conjugates, the lit length, and the focus of the LED image on the slit unchanged. It converges each cone by about 5 %, so the iris is set to about Ø10.0 mm, a cone of ±1.91° at the slit, for the cone to reach the mirror's edge from every point of the lit slit.

Defocus in either direction underfills the ±1.82° cone: iris-edge rays then trace back beyond the emitter edge, darkening the mirror edge and shortening the uniformly lit slit length. With the LED at the condenser focus (12 mm, collimated) the emitter cannot fill the cone even at the slit center.

The iris-to-slit distance follows from the stack (§6.1): the leaves of the SM1D12 lie 9.4 mm ahead of the lens's convex vertex. That is 0.7 mm for the vertex's depth inside the SM1V05's open end (the 12.0 mm lens in its 12.7 mm seat), 8.38 mm for the SM1L03 (the iris's thread shoulder seats on its open end), and 0.36 mm for the leaves' midplane past that shoulder. The last is from the vendor model, whose 10 leaves span 0.76 mm before to 1.47 mm past the shoulder; the vendor drawing does not locate them. With 159 mm from the vertex to the slit, the leaves are 149.6 mm from the slit.

Using ACL2520U-A catalog values (EFL 20.1 mm, BFL 12.0 mm; object-side principal plane about 8.1 mm inside the plano face) and 159 mm from the convex vertex to the slit, the optimum emitter-to-plano-face gap is about 14.9 mm. The optimum moves only about 0.5–0.8 mm for 30 mm of slit position either way.

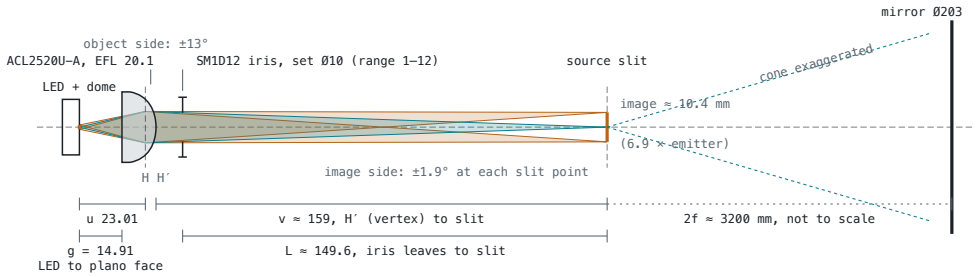


Fig. 8. Condenser layout unfolded (mirror reflection removed). Axial and transverse scales match from the LED to the slit; the slit-to-mirror span and the cone beyond the slit are not to scale. Teal: the on-axis emitter point; amber: the two emitter edges

The prescription, with the basis of each value:

Element	Value	Basis
Emitter (apparent, through dome)	≈ 1.5 mm (green)	Estimate: ~1 mm die scaled from the OSRAM outline × 1.4–1.5 dome; white 519A phosphor size not recorded
Emitter to plano face	14.91 mm (13–17 usable)	Calculated optimum for 159 mm vertex-to-slit
ACL2520U-A condenser	EFL 20.1, BFL 12.0, NA 0.60, CA > 22.5	Catalog; object-side principal plane H at EFL − BFL = 8.1 mm inside the plano face
Conjugates	u 23.01, v 159, m ≈ 6.9	Calculated from 1/u + 1/v = 1/f; u from H, v from the image-side principal plane H′ at the convex vertex
SM1D12 iris	Ø1–12 range; ≈ Ø10.0 set	Catalog range; setting calculated as $D = 2 \cdot L \cdot \tan\theta$ at L = 149.6 mm, θ = 1.91°, the cone that the field lens converges onto the mirror's ±1.82° from every point of the lit slit
Iris leaves to slit	149.6 mm (leaves 9.4 mm ahead of the vertex)	The distance at which the field lens images the iris onto the mirror; the leaves' place in the stack: vertex 0.7 mm inside the SM1V05 open end, SM1L03 8.38 mm, leaf midplane 0.36 mm past the SM1D12 shoulder (vendor model); 159 mm vertex-to-slit
Source slit	0.10–0.30 wide, 10.4 mm lit length	§7
Field lens	LA1433-A, Ø1 in, f 149.5; plano face ≈ 5 mm ahead of the slit	Catalog; the slit is placed for this focal length to image the iris onto the mirror (above); §7.9
Mirror	f 1600; slit at 2f = 3200 mm, ±101.5 mm, ±1.82°	Catalog D203F1600; slit at the center of curvature (§3.2); ±2.0° gives slight overfill

Each emitter point sends a cone of about ±13° into the lens, and its image point receives a cone of about ±1.9°. The iris sits in the converging space and sets that angle, so all emitter points share it.

Moving the LED re-establishes the conjugates with the same lens: the slit moves to stay in focus and the iris is resized to keep the cone that fills the mirror. The lit length is the apparent emitter (≈ 1.5 mm) times m, and the flux through the slit scales with it:

Gap g	m	v (vertex to slit)	Iris needed (D = 2 L tan θ, θ = 1.82–2.0°)	Lit length	Result
14 mm	10.1×	≈ 222 mm	≈ 13.5–15 mm	≈ 15 mm	Iris cannot open far enough (12 mm maximum); image longer than the 10 mm needed
14.91 mm	6.9×	≈ 159 mm	9.5–10.4 mm	≈ 10.4 mm	Design point; the slit distance the field lens needs, and the image covers the lit length (green)
16 mm	5.0×	≈ 121 mm	≈ 7–8 mm	≈ 7.5 mm	About 25% less light through the slit; the iris is then about 112 mm from the slit, so the field lens (matched to about 150 mm) leaves the lit end's cone about 39 mm off the mirror's center; a lens of about 115 mm would center it

Focus

The LED module seats against the rear face of the SMR1/M (§6.1), which fixes the emitter, and the SM1V05 sets the lens position by how far it is threaded into the SMR1/M. Each millimeter of added engagement moves the lens a millimeter closer to the emitter, so the engagement range sets the range of reachable emitter-to-plano gaps. Engagement runs from the Thorlabs minimum of 2.79 mm to about 5.7–5.8 mm, where the sleeve end comes within 0.3 mm of the LED board. Turning the SM1V05 moves the lens and iris together, but the resulting few-millimeter iris shift is negligible against the 150 mm iris-to-slit spacing, so the iris setting is unaffected. The gap is measured to the emitting surface, not the dome top.

The emitter heights above the cap face are calculated from the measured board thicknesses, about 0.07 mm of solder, and the apparent emitter heights through the LED domes (from the manufacturer outline drawings: OSLON about 0.45 mm, 519A about 0.75 mm). With them, each module reaches the following gap range over the engagement range:

Module	Emitter height above cap face	Emitter-to-plano gap range	SM1V05 engagement at 14.91 mm gap
Green	about 2.1 mm	about 13.2–16.2 mm	about 4.1 mm
White	about 2.3 mm	about 12.9–16.0 mm	about 3.9 mm

Focus margin about the 14.91 mm optimum gap is about 1.3 mm (long-gap side) and 1.7 mm (short-gap side) for the green module, and about 1.1 mm and 2.0 mm for the white module. The white module's long-gap margin is the smallest, its travel limited by the SM1V05 minimum engagement. **Provisional:** the white module's long-gap margin is to be confirmed on the build. If it cannot reach a sharp emitter image on the slit, space its cap flange off the SMR1/M face with a thin shim; each 0.1 mm of shim adds about 0.1 mm of long-gap margin. The shim moves the whole module away from the lens and keeps the flange as the axial stop; a shim under the star board would instead shorten the gap and make the shortfall worse.

To set up, thread the SM1V05 in to the engagement above for the module in use, set the iris to about Ø10.0 mm, and place the slit head about 150 mm from the iris leaves (159 mm from the lens vertex). The blades of the stage-based slit head cannot be seen, so swap the printed focus target in for the head (§7.7), with the field lens station

slid clear (§7.9), turn the SM1V05 until the emitter image on the target is sharp and centered, then lock the SM1NT, restore the slit head and the field lens, and open the slit. The two modules focus about 0.2 mm apart in engagement, about one-third of a turn of the SM1 thread; an index mark per module on the SM1V05 is optional.

In transport, the condenser/iris assembly is protected from dust and scratches by a Thorlabs SM1CP1 internally threaded end cap and an SM1EC2 snap-on dust cap.

7. Source slit

The source slit is a pair of razor blades, one on the table of a purchased 40 × 40 mm linear stage and one on a printed pedestal, with the stage's micrometer setting the slit width. The head is mounted for continuous rotation about the optical axis in an SM1RC/M on the common post and rail shoe (§5). A baffle behind it on the source side is clamped on the head's spigot and turns with it (§7.8), a field lens on a post of its own ahead of the blades makes the mirror illumination uniform (§7.9), and a printed focus target stands in for the head to focus the LED image (§7.7).

The model is `src/schlieren/parts/slit_stage.py` (`uv run slit-stage`, with `--show` for the viewer assembly).

The slit needs one adjustment, its width, and no interchangeable elements, so a purchased stage with a micrometer serves where the cutoff uses a carriage and cassettes (§8).

7.1 Blades and slit widths

The slit uses two Stanley 11-515 backed single-edge scraper blades.

Measured physical blade dimensions:

- length: 1.537 in / 39.04 mm
- width, cutting edge to the outside of the fold: 0.773 in / 19.64 mm
- plate thickness: 0.010 in / 0.254 mm
- fold: the back edge is folded over both faces, 0.277 in / 7.04 mm deep on each, 0.040 in / 1.016 mm thick across the fold
- bevel on both faces, breaking 0.030 in / 0.76 mm back from the cutting edge.

The blade has a slot at mid-length and a notch in each end, approximated in the CAD from a photograph (about ±0.3 mm); the slit and cassette fixtures do not rely on them. The CAD model is `parts/razor_blade.py`.

Expected operating slit widths are 0.10–0.30 mm, starting around 0.15–0.20 mm.

Each blade lies on its fold, flat on its seat, the stage table or the pedestal, cutting edge toward the slit and overhanging the seat edge by 2.9 mm. The fold's outer layers stand 0.38 mm proud of the plate's faces, so with the fold's outer layer on the seat the plate is parallel to it, 0.38 mm above it and clear of its edge, and the cutting edge's mid-plane is 0.51 mm above the seat plane. No blade pockets, holes, or spine compensation are used. The optically important geometry is the gap and the parallelism of the two cutting edges: the two blades are identical and seated alike, so their edges lie at the same height, collinear were the slit closed, to within the difference of the two folds' thicknesses. The 12.6 mm of plate in front of the fold is a stiff cantilever: calculated, its own weight deflects it by about 0.2 µm and its first resonance is about 1.3 kHz.

7.2 Architecture

The head is a stage plate carrying the stage and a pedestal:

1. stage plate: a 6 mm printed plate with the spigot and shoulder for the SM1RC/M;
2. stage: a 40 × 40 mm linear stage with a micrometer drive (Amazon B0B7RVW3SX, an SF Technology SEMX40-AS type, ±6.5 mm of travel), bolted to the plate front; its table carries the movable blade;
3. pedestal: a printed block on the plate carrying the stationary blade at the same height as the table.

Local frame, at rotation 0: x along the slit, z the adjustment axis with the micrometer toward +z, y along the optical axis toward the mirror; the origin is the slit center in the blade-seat plane. The blade-seat plane (the table top and the pedestal top) is the front of the head; the stage's 20 mm thickness puts it 20 mm ahead of the plate front. In

this section "front" and "back" refer to the head, not to the rail direction.

The head's front faces the mirror. The light comes from the source side through the baffle and the plate's Ø24 mm bore and passes the stage's edge to reach the slit from behind, so the spigot, ring, and post are on the light-source side of the blades. The source cone is ±2° about the chief ray and the lit slit length extends to ±7.8 mm (§6.3); the stage and pedestal edges keep 1.65 mm from the cone.

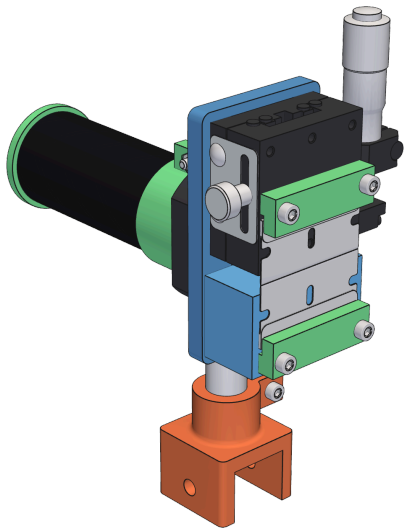


Fig. 9. Stage slit head and baffle on its post and rail shoe, seen from the mirror side: the movable blade on the stage table, the stationary blade on the pedestal, a toe bar over each, and the micrometer up

7.3 Slit width

- drive: the stage's micrometer, moving the table and so the movable blade; travel ±6.5 mm (manufacturer);
- the slit is 0.2 mm in the model, within the 0.10–0.30 mm working range of §7.1.

There is no centering stage. Only the movable blade moves, so a width change shifts the slit center by half the change. The iris is about 150 mm before the slit, so moving the slit by d turns the illumination cone by d/150; at the mirror the beam moves about 22 × d. A 0.1 mm width change therefore moves the slit center 0.05 mm and the beam about 1 mm at the mirror, which needs no recentering (the cutoff is readjusted for any width change in any case). With the field lens (§7.9) in place, imaging the iris onto the mirror, the beam stays centered there.

Slit setup:

1. with both toe bars loose, bring the movable blade to the pedestal blade with the micrometer;
2. seat both blades and clamp them progressively (§7.4);
3. set the width against a true-metric feeler gauge between the cutting edges, pressing gently along their length;
4. thereafter set the width with the micrometer from that reference. Re-check parallelism with the gauge after large width changes.

7.4 Blade clamping

Each blade is held on its seat by a printed PLA toe bar. One bar sits on the stage table and one on the pedestal; each is screwed down at its two ends into the stage's M3 table holes (second row, 16 mm pitch) or into M3 heat-set inserts (McMaster 97171A310) in the pedestal. The bar's finger runs the length of the blade and presses the rear 3 mm of the blade's fold flat onto the seat through a strip of 1/32 in EPDM (McMaster 9852N37, the stock of §4.3). The finger does not touch the plate or the cutting edge, and no blade is drilled.

The clamp holds the blade only against its own weight and handling and alignment loads, a small force, and the fold is stiff. The toe's squeeze on the EPDM pad is 0.3 mm, which puts the finger's underside 1.51 mm above the seat plane: the fold's 1.02 mm plus the compressed pad. The finger's slit-side face stands 16.2 mm from the optical axis, 1 mm clear of the Ø30.5 mm body of the field-lens tube (§7.9), which stands between the two fingers over the axis; the head turns about that axis, so the fingers clear the tube at every angle. The common McMaster 91292A114 M3 × 12 mm screw is used.

7.5 Rotation mount and clearances

The stage bolts to the plate front with four M3 screws on its 32 mm base-hole pitch, with captive M3 nuts (the common 91828A211) in pockets in the plate rear. The pedestal is screwed to the plate from the rear into two M3 heat-set inserts in the pedestal (97171A310).

The plate carries a hollow Ø30.48 mm (SM1 tube OD) spigot with a Ø24 mm light bore, clamped by the SM1RC/M split ring (Ø30.7 mm bore, 10.16 mm thick, M4 locking screw across the split). A Ø34 mm shoulder on the spigot is the ring's axial stop: the ring seats against it, which places the plate 3.5 mm in front of the ring face. Do not clamp the ring away from the shoulder. The nut pockets reach inside the spigot's radius, so the spigot is restored behind the shoulder and roofs them, and the ring grips a full cylinder.

The spigot runs the ring's full thickness and protrudes 12 mm past its source face, where the baffle's clamp (§7.8) grips it; the ring clamps nothing else. The SM1RC/M keeps the 22.1 mm post-top-to-axis dimension (§5.1). Loosening the ring rotates the whole head about the optical axis:

- rotation 0: horizontal slit, micrometer up;
- ±90°: vertical slit, micrometer sideways. With the micrometer toward the imaging rail it reaches the cutoff carriage's base plate unless the cutoff is staggered behind the slit by about 16 mm or more (the 30–40 mm starting stagger of §3.3 clears it by about 20 mm); turned outboard it clears at any stagger;
- small rotations: slit-to-cutoff parallelism trim.

The rotation sweep clears the post, ring, and rail shoe in the tests.

7.6 Fabrication and fit

Printed parts: the stage plate, the pedestal, and two toe bars.

- Plate: print front face down, spigot up; no supports.
- Pedestal: print top face down, so the blade seat is the first layer.
- Toe bars: print flat, heel side down.
- The spigot is printed at the SM1 tube nominal, which fits the SM1RC/M.

Hardware: the stage, 4 × M3 × 10 mm screws (91292A113, stage to plate), 4 × M3 nuts, 2 × M3 × 12 mm screws (pedestal to plate), 4 × M3 × 12 mm screws (toe bars), 4 × M3 heat-set inserts, 2 strips of EPDM.

Provisional:

- The stage's dimensions are the manufacturer's (the table layout, 20 mm thickness, ±6.5 mm travel), not measurements of the stage on hand. The model shows Misumi's XCRS40, which has the same table and base layout and a solid CAD model. The base-hole pitch, the table's tapped holes, the micrometer reach, and the base flange thickness are to be measured, and the base mounting holes (four M3, counterbored) are assumed reachable from the table side.
- The blade overhang and the toe bar's squeeze on the EPDM pad (0.3 mm) are first-print values, as is how flat the stamped folds lie on the seats and under the pad (§7.1, §7.4).
- The M3 × 10 mm stage screws leave about 1 mm of thread behind the plate, 2.5 mm short of the ring face, with the thread still showing past the nut; the length rests on the stage's base flange thickness, which is not yet measured.

7.7 Slit focus target

The emitter image is focused on the slit plane (§6.3), but the blades of the head sit deep in the stage and cannot be seen. A printed focus target swaps in for the stage plate on the same SM1RC/M ring: the same shoulder, ring gap, and spigot grip, then a hollow barrel that widens to a Ø40 mm rim whose face is the plane of the blades' cutting edges, 0.51 mm ahead of the blade-seat plane (§7.1). The Ø24 mm bore tapers in at 45° to Ø20 mm at the rim face, which gives the paper or glass a wide seat and the insert recesses a thicker wall; the beam there is about Ø11 mm. The target's spigot ends flush with the ring's source face, so the baffle (§7.8) is taken off the plate's spigot first: loosen its clamp screw and slide it off. The focus models do not show it. The models are `src/schlieren/parts/slit_focus_target.py` (`uv run slit-focus-target`, with `--show` for the paper variant and `--show --glass` for the ground-glass one).

A sheet of tracing paper lies on the rim and a printed Ø40 × 4 mm retaining ring clamps it. The ring's Ø20 mm window matches the seat bore at the rim face, so the ring clamps the full width of the rim's annulus; it passes the whole beam and stays within the plate's Ø24 mm bore. The window's mirror-side edge has a 1.5 mm × 45° bevel for a wider viewing angle; the 2.5 mm land behind it clamps the paper, and the screw heads bear outside the bevel. Four M3 × 10 mm socket-head screws (91292A113) on a Ø32 mm circle run into heat-set inserts (97171A310) in the rim. The LED image forms on the paper's source-facing side and is read through the paper from the mirror side, so the turn of the SM1V05 can be judged by eye before the head is fitted. The rim face is the datum, so the paper's thickness (not yet measured; the model assumes 0.15 mm) offsets the focus plane from the blades' edge plane by that much. For handling, the rim's back edge and the ring's front edge, both Ø40 mm, have a 0.5 mm chamfer; the mating faces keep sharp edges.

A variant replaces the paper with a Ø1 in ground-glass disc (Thorlabs DG10 series, catalog 2.0 ± 0.2 mm thick). The printed fixture is the same; only the retaining ring differs. The disc sits in a Ø25.7 mm pocket in the ring's rear face, ground face toward the source, lying on the rim face as the paper does, so the imaging surface is the blades' edge plane whatever the glass thickness (the paper's own thickness aside). A gasket cut from 1/16 in Durometer 30A neoprene sheet, between the glass's front face and the pocket ceiling, pushes the glass back onto the rim and absorbs the thickness spread. The disc has a 0.2 mm × 45° bevel on both edges (Thorlabs model; the drawing does not dimension it), so its flat faces are Ø25.0 mm, and the gasket is cut to that. The ring is 4.65 mm thick (3.15 mm pocket, 1.5 mm lip) with a 1 mm × 45° bevel on the window, and takes the same M3 × 10 mm screws. The glass is set on the rim, then the gasket, then the ring over both; the flat seats on 2.5 mm of the rim, outside the Ø20 mm seat bore. **Provisional:** the pocket clearance is to be set from a printed coupon, and the grit (DG10-600 to start) is to be confirmed on the bench.

7.8 Baffle

A baffle sits on the source side of the slit head, on the optical axis: one printed part, a clamp ring on the plate's spigot followed by a lattice tube that carries a matte black foil wrap. It is built from the spigot out, so the posts, shoes, and optical-mechanical rings of the other stations are not needed.

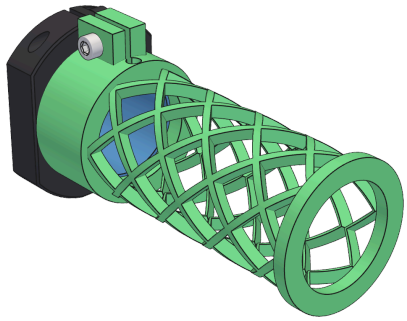


Fig. 10. Baffle on the plate's spigot, seen from the source side, without its foil wrap: clamp ring, slit relief hole, net, far collar

Motivation. The condenser focuses the LED image, about 10 mm across, on the blades (§6.3), and a slit of 0.10–0.30 mm passes only a small fraction of it. The blades return the remainder, and the head's lit surfaces scatter some of it. That light goes back toward the source as a broad cloud, strikes the bare post, the source hardware, and the table behind the head, and is forwarded by them, glossy surface to glossy surface, toward the mirror and the operator at the imaging station. The cloud lights the room and adds veiling glare to the image. A tube immediately behind the head confines the return to the direction of the LED: it travels back through the bore toward the iris, and the rest meets the black walls and vanes.

Beam. The baffle bounds the light but does not stop down the beam. The beam on the source side of the slit lies between the iris aperture (about Ø10 mm, §6.3) and the lit slit length (to ±7.8 mm), so it is about Ø16 mm over the whole baffle, with the ±2° cone about the chief ray inside that envelope. The Ø25 mm bore leaves about 4.5 mm of radius all around, and is no narrower than the plate's Ø24 mm spigot bore, so no alignment tolerance clips the beam, and the length is set by the clamp and the distance to the iris, not by the beam. It ends 81 mm from the ring's source face, about 21 mm short of the iris housing. At the slit end the beam is bounded by the head instead, not the baffle: the stage and pedestal edges keep 1.65 mm from the cone (§7.2).

Mount. The plate's spigot protrudes 12 mm past the SM1RC/M's source face. The baffle's clamp ring, 12.5 mm long, fits over that end and butts the ring's face, which sets its axial position. The clamp is built as the rail shoe's collar is (§5.2): a Ø30.68 mm bore (0.20 mm diametral clearance on the spigot), a 1.5 mm split along the top, and ears with one M3 × 12 mm screw (91292A114) and a captured M3 nut (91828A211), 3.5 mm ear on the screw side and 4.5 mm on the nut side. The slit ends in a Ø3.5 mm round relief hole in the solid collar behind the clamp bore, so the ring closes on the spigot without cracking at the slit's end. The bore stops at a shoulder 0.5 mm short of the spigot's end. The ears stand about 24 mm above the axis at rotation 0 and turn with the head, which they clear through its working range. The baffle needs no precision of its own: the spigot, held by the head's ring, sets its axis, and the bore's radial margin over the beam (above) absorbs the clearance of the two printed fits.

Net. Behind the collar the baffle is a lattice tube, Ø31 mm outside, with staggered diamond windows cut through its wall so that the ligaments between them (1.4 mm wide) run at 45° to the axis. That lets it print on end without supports and be stiff enough to support its own length and the foil. The ligaments are 3 mm deep, standing into the bore as vanes, which leaves a Ø25 mm bore. A stray ray that strikes a vane's flank is trapped there, where it would otherwise slide along a bare wall. Six windows run around the tube, in nine rows staggered by half a window; the end rows are centered on the collars' faces, so they are triangles. A solid ring collar closes the net's far end.

Foil and finish. The net's inside is painted with flat black spray paint (a rattle can) before the foil goes on; the vanes' flanks and the bore's inside surface are the baffle's absorbing surfaces, and the clamp's bore and the spigot are kept free of paint. The foil is Rosco Photofoil matte black aluminum wrap (BOM SHP-022), wrapped once or twice around the outside of the net between the two collars, matte side in, so the windows show foil; a strip of tape across the seam holds it. The collars stand 1.8 mm above the net's surface, which guides the foil's edges and leaves it flush with them. Wrapping the outside avoids fitting a liner into the bore.

Fabrication. Print on end, clamp end down, in PLA, without supports; the ears and split lie along the layers. The model is `src/schlieren/parts/baffle.py` (`uv run baffle`, with `--show` for the viewer assembly), and `slit_stage.py` shows it with the head.

Provisional: the clamp's bore clearance and split gap are first-print values, to be settled with a coupon print; the net's stiffness as a cantilever of its own length, the vanes' depth, and the effect of the painted vanes on the glare are to be confirmed on the bench.

7.9 Field lens

The field lens makes every slit point's cone fill the mirror alike (§6.3), and the slit stands at the distance from the condenser at which the lens's focal length images the iris onto the mirror. It is a Thorlabs LA1433-A, a Ø1 in N-BK7 plano-convex lens with f = 149.5 mm, on a post of its own ahead of the blades, not on the slit head. Three reasons: the lens's alignment to the optical axis rests on the machined metal parts of the common post stack (§5) and not on a printed dimension; the lens does not turn with the head, so a decentering does not swing the illumination around when the head is rotated; and the head's toe fingers only stop short of the tube (§7.4). The model is `src/schlieren/parts/field_lens.py` (`uv run field-lens`, with `--show` for the viewer assembly), and `uv run optical-head --no-field-lens` shows the head without it.

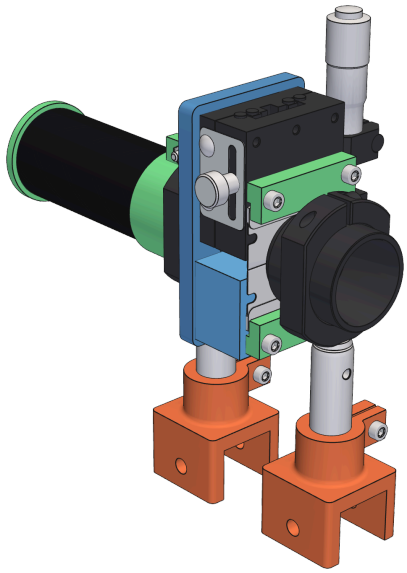


Fig. 11. Field lens station ahead of the slit head, with the head's post, seen from the mirror side

Stack. The lens sits with its plano face on the internal lip of a Thorlabs SM1L10 lens tube, 3.81 mm from the tube's thread end, and its convex face toward the mirror. The SM1RR retaining ring supplied with the tube holds it from the open end; seat the ring with the optical spanner wrench. The tube is clamped by its full-diameter section in a Thorlabs SM1RC/M split ring on a TR50/M post on the common shoe, the same stack as every other station, so the lens is at the 22.1 mm post-top to axis dimension of the common post stack.

Position. The tube's thread end stands 0.5 mm above the tops of the blades' folds, between the toe fingers, whose slit-side faces stand 1 mm clear of the tube's body (§7.4). The lens's plano face is 3.81 mm in from the thread end, about 5 mm ahead of the blade-seat plane (5.3 mm in the model). The tube is long enough to carry the lens from a clamp ring that stands clear of the toe hardware, the reason for a tube in a split ring rather than a one-piece lens mount: the toe bars (to 6 mm ahead of the blades) and their screw heads (to 9 mm) sweep through the region around the axis. The screw heads set the post's position: they pass the post 2 mm clear (§7.5) at the head angle that puts one over it. The ring's slit-side face is about 12 mm ahead of the blade plane and the post axis about 17 mm ahead, 52.0 mm from the slit head's post, which leaves about 22 mm of rail between the shoes. The station along the rail is a setup value, set by sliding the shoe, and the tube's position in its ring sets the lens's distance from the slit; neither is critical, since 2 mm either way moves the footprint from the end of the brightest portion on the mirror by about 1.4 mm (about 2.1 mm from the shoulder's edge). The slit lies inside the lens's focal length, so the mirror sees a virtual slit image about 5 % larger than the slit and about 0.4 mm behind it, and the returned slit image at the cutoff is that much larger (§9.3). The beam at the lens is about 17 mm long, inside the tube's Ø24 mm bore at the lip and the retaining ring's Ø22.9 mm aperture.

Alignment. A lens decentered by δ shifts the whole illumination on the mirror by about 21 δ; with the lens centered, every slit point's cone centers on the mirror. Mirror height corrects the component of the shift across the slit, as for the slit-width shift of §7.3; the component along the slit is limited by the post stack's alignment. Opening the iris from Ø10.0 mm to its Ø12 mm maximum widens each footprint by about 20 mm at the mirror, enough for a lens error of about 1 mm.

Without the field lens. To compare with and without it, loosen the SM1RC/M's M4 screw and slide the tube out of the ring toward the mirror; the empty ring and post stand clear of the beam. Slide the tube back to restore the lens, to within about 2 mm of its position. Slide the whole station away from the slit when the head is removed or the focus target is fitted (§7.7): the tube's thread end stands between the toe fingers, over the blades, where the target's rim and ring go, and the paper is read from the mirror side. The lens does not change the focus of the LED image on the slit plane.

Provisional: the lens's position and the iris setting rest on the iris-to-slit distance of about 150 mm (§6.3), which is calculated from the stack and not measured; both are to be confirmed from a flat-field photograph of the mirror illumination with the lens in place and removed.

8. Cutoff

The cutoff sits at the returned slit-image plane on the imaging rail. It is a rotating carriage (§8.1–8.5) holding interchangeable 64 mm cassettes (§8.6–8.7), which carry the cutoff element: a knife edge, a fine wire (§8.8), or a color filter (§8.9). The carriage model is `src/schlieren/parts/carriage.py`; the cassette blank and clamp bars are modeled in `src/schlieren/parts/cassette.py`.

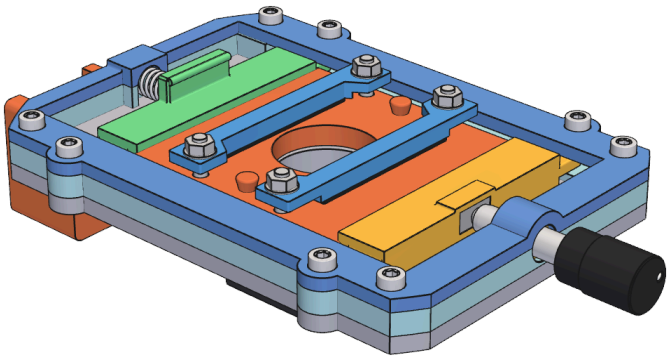


Fig. 12. Cutoff carriage with a seated cassette: guide frame, keeper frame and its retaining screws, driven and spring plungers, and the FAS100 fine adjuster

8.1 Spigot and rotation interface

The carriage mounts in 1 × Thorlabs SM1RC/M through a hollow printed spigot integral with the base plate, the same interface as the slit head's plate (§7.5):

- Ø30.48 mm spigot (SM1 tube OD nominal), clamped directly by the SM1RC/M split ring (Ø30.7 mm bore, 10.16 mm thick, M4 locking screw across the split); the spigot end is flush with the ring's rear face;
- Ø24 mm light bore through the spigot, matching the plate aperture;
- Ø34 mm shoulder as the ring's axial stop: the ring seats against it, which places the plate back 12.0 mm in front of the ring face. Do not clamp the ring away from the shoulder.

The 12.0 mm ring gap is set by the rail shoe, not the post. At nominal zero the plate reaches about 63.5 mm below the optical axis, past the shoe top (about 52.4 mm below), so the plate back must clear the shoe end (15 mm from the post axis): calculated clearance 2.1 mm to the shoe and 10.7 mm to the post body, at every rotation. The shoulder clears the post top by 5.1 mm.

The SM1RC/M provides the rotational bearing and lock directly on the spigot: when loosened, the carriage rotates on its spigot. Nothing retains the spigot axially while the ring is loose, so support the carriage when rotating it.

Required carriage rotation:

- at least ±90°
- preferably ±100–120°

Define nominal zero as:

- cutoff line horizontal;
- fine-adjust translation vertical;
- fine adjuster pointing upward.

The cassette-loading side of the carriage faces the mirror, so cassettes are loaded and retracted from the forward side, away from the telephoto. The spigot, split ring, and post are therefore aft of the base plate; the ring's rear face, 22.2 mm behind the plate back, is the rearmost part of the station on the optical axis and is what the telephoto front must clear.

Fore/aft optical positioning is done by moving the rail shoe.

Rotation envelope: the 2020 rail runs beneath the carriage at every rail position, 72.35 mm below the axis. The base plate, guide frame, and keeper are trimmed to a 70.35 mm radius about the optical axis (the rail top less 2 mm clearance). Only the outer plate corners are removed; the plate is otherwise 86 mm wide, widened to 94 mm at the side keeper-screw lugs (§8.4). Calculated from the CAD, the carriage with its FAS100 rotates clear of the rail over ±150°; only the FAS100 knob reaches the rail, near ±170–180°. Nothing projects below the plate at nominal zero (§8.5).

8.2 Spigot print and fit

The base plate is printed deck down with the spigot up, so the spigot and shoulder need no supports; the guide frame above the deck is a separate print (the base/frame split exists for this). The three datum-pin shanks (§8.4) lie outside the shoulder and are flush-cut on the plate back.

The spigot is printed at the SM1 tube nominal, which fits the SM1RC/M without adjustment.

8.3 Fine adjustment

The carriage uses:

- 1 × Thorlabs FAS100, 1/4 in-80 × 1.00 in
- 1 × McMaster 98625A950 brass 1/4 in-80 insert
- 1 × McMaster 2006N292 compression spring

Insert geometry is from the McMaster 98625A950 drawing: internal thread 1/4 in-80; body outside diameter 0.313 in (7.950 mm); overall axial length 0.313 in (7.950 mm); under-flange/body length 0.298 in (7.569 mm); flange outside diameter 0.352 in (8.941 mm); flange thickness 0.010 in (0.254 mm); specified drill-bit size 0.313 in (7.950 mm); and minimum material thickness 0.298 in (7.569 mm).

The insert bore is nominally 0.313 in (7.950 mm), matching the drawing. The printed bore is a pilot, finished after printing with a 5/16 in (0.3125 in) drill rather than relying on FDM accuracy, to a press fit. The 9.3 mm insert support exceeds the drawing's 0.298 in (7.569 mm) minimum material thickness, leaving 1.7 mm of PLA beyond the end of the insert body (1.6 mm past the entry chamfer) and 3 mm of keeper crown over the bore. The spring plunger loads the FAS100 tip, so the screw pushes the insert outboard, toward the knob. The flange is therefore on the inner (cassette-side) face of the adjuster block, with the body running outboard: the reaction bears the flange against the block instead of relying on the press fit. The flange stands 0.254 mm proud of the flat block face; the plungers stop about 9 mm short of it. The insert is pressed in from the cassette side before the plungers are installed, with the flange bottoming on the face.

The adjuster axis is 8.77 mm above the plate back: the 8.94 mm flange clears the deck by 0.3 mm, so the flange seat is a plain face and the plate is not notched. The spring axis (§8.5) is the same line, so both plungers load the cassette along one axis.

FAS100 / insert travel envelope for CAD: the FAS100 uses a 1.00 in (25.4 mm) long 1/4 in-80 adjuster screw. The 80 TPI pitch gives 0.0125 in (0.3175 mm) axial motion per revolution. Using the insert's 0.313 in overall axial envelope as a conservative full-engagement allowance leaves approximately 0.687 in (17.45 mm) of theoretical screw motion. That is a geometric clearance estimate, not a published FAS100 travel specification. The carriage's working translation is ±5 mm / 10 mm total, leaving useful adjuster margin before either end stop.

The FAS100 ball tip does not bear directly on printed PLA. The driven plunger carries a flat bearing insert in a close-fitting printed pocket: a nickel-plated N52 NdFeB bar magnet, 10 × 5 × 2 mm (Amazon B0DMCY4FN1). The plated face is a hard, wear-resistant bearing surface, and the magnet is functional: its attraction to the FAS100 ball tip helps retain the driven plunger on the adjuster screw during cassette changes, when no cassette is loading the plunger against the screw. The bearing insert has its rear face fully supported by PLA, with a small cyanoacrylate (CA) or epoxy tack for retention. The FAS100 ball contacts the broad plated face rather than bare PLA. The lens cradle (§9.5) uses the same bearing insert; in the cradle the magnet's pull on the steel screw tips likewise adds some preload (§9.8).

The plunger top is held near 12.0 mm by the cassette clamp-bar relief, which leaves only 0.6 mm of roof over the pocket (top at 11.37 mm). A local crown on the driven plunger over the pocket, 5 mm long from the outer face and 1.5 mm wider than the pocket on each side, restores a 1.3 mm roof (top at 12.67 mm). It is at the outboard end, clear of the clamp bars; it is blended into the plunger top by a 0.4 mm root fillet on its sides and inboard end.

8.4 Cassette seating

Cassette motion is controlled by opposed plungers:

- fine adjuster on one side;
- spring-loaded plunger opposite;
- three fixed rear datum contacts on the carriage define the cassette optical plane;
- shallow cassette/plunger bevels generate axial seating force against the datum contacts.

The three rear datum contacts are McMaster 97936A109 brass escutcheon pins / decorative finishing nails, used head-first as small domed sliding contacts. Each pin is approximately 3/8 in long, with a 0.050 in (1.27 mm) smooth shank and a measured 0.093 in (2.36 mm) diameter, 0.038 in (0.97 mm) high domed head; the head height sets the cassette's standoff from the datum face.

Mount each pin through a through-hole in the printed carriage with the domed brass head on the cassette side. The holes are modeled Ø1.75 mm: the measured 0.050 in (1.27 mm) shank, plus a 0.38 mm diametral allowance for the undersize of printed holes, plus 0.1 mm fit clearance; the holes take the shanks as a slip fit, so CA (below) is their only retention. If a shank binds, finish the hole with a #54 (0.055 in) drill. The hole fit only locates the shank; the head sets the datum. Seat the underside of the head firmly against the carriage datum face so datum height is established mechanically rather than by adhesive thickness. Wick a small amount of CA around the shank from the back for retention, then flush-cut and dress the projecting shank at the back of the carriage. Keep CA out from beneath the head/contact surface.

The three pin heads form a broad triangular support pattern around the optical aperture. The cassette rear face is the sliding datum surface and provides unobstructed swept tracks over the three domed contacts throughout the fine-adjust travel. The pin coordinates, set in the carriage CAD, keep every datum contact out of the swept track of the cassette's rear M3 clamp-screw countersinks over the full travel, so the datum tracks stay on uninterrupted printed rear-face material at every cassette position.

The plunger bodies are constrained to the intended translation axis by ears/tabs riding in guide slots in the carriage. The slots prevent unwanted lateral movement or rotation while permitting the required axial plunger travel.

The ears run with only 0.15 mm of clearance below and 0.25 mm above, so their exposed edges are broken against binding: a 0.4 mm chamfer on the bottom edges, which take the printed first-layer squash, a 0.5 mm fillet on the top edges, and 0.6 mm fillets on the outer vertical corners (larger than the two edge breaks they meet). Where each ear leaves the plunger body wall, a 0.8 mm concave fillet on both the top and the underside relieves the stress: the underside root is on the tension side under the seating reaction (about 10 N per ear, a few MPa nominal). It lies at x = 29.0–29.8 mm in the carriage frame (origin on the optical axis, y along the travel, +y toward the FAS100), clear of the guide-rail floor (x ≥ 32.3 mm) and the keeper opening edge (x = 33 mm).

The plunger bodies themselves run 0.3 mm above the base-plate deck, so their underside perimeter carries a 0.4 mm chamfer, a lead-in over deck seams and a relief for first-layer squash. On the spring plunger it continues around the feet of the spring cup, including the edges of the pocket's opening toward the deck; the cassette-side bevels and lips, which generate the seating force, are left as designed.

The plungers need an assembly path into their guide slots: both longitudinal ends of each guide rail are occupied, by the FAS100 adjuster support on one side and the spring end wall on the other. The guide rails are therefore open from the cassette-loading side; the plungers drop onto them, and a separate printed keeper frame closes only the guide-ear regions, completing the slots.

The keeper is a perimeter frame, not a solid lid: the central upper aperture must remain open for cassette insertion and removal. Its side members span the plunger guide rails, completing the slots, and retain the guide ears against upward escape while the printed rail walls carry the lateral guidance loads. The frame may be locally scalloped or notched where the FAS100 adjuster axis intersects its envelope, provided the material directly over the swept guide-ear paths and the load paths to the fasteners remain intact. The keeper's spring-end member also carries the roof of the end-wall spring cup (§8.5).

Keeper retention is mechanical, with eight fasteners per carriage. Each is the common McMaster 91292A114 M3 × 0.5 × 12 mm socket-head screw with a McMaster 91828A211 full-height M3 × 0.5 hex nut (5.5 mm across flats × 2.4 mm high), captive in a hex pocket opening from the plate back. Screw heads and nuts bear directly on the PLA, without washers; tighten only to a light snug, since the joint needs little clamp and a firm hex-key torque can crush the bearing face.

- Four end screws, at (±38.5, −53) and (±38.5, +52) mm in the carriage frame. Each sits on solid fill closing its guide slot from 1 mm beyond the outermost ear position to the plate end. The fill is also the plungers' end stop: 1 mm past full loading retraction for the spring plunger and past +5 mm travel for the driven plunger. These positions keep each nut pocket at least 1 mm inside the rotation envelope (§8.1).
- Four side screws, at (±42, ±37) mm, beside the middle of each plunger's ear travel. Each sits in a Ø10 mm round lug that widens the base, frame, and keeper locally to 94 mm, clipped at the guide-rail wall.

The side screws carry the upward ear reactions from the bevel wedge action (sized below) into the frame close to where they arise. A first-order beam estimate, with each keeper side member simply supported between adjacent screws, gives at most about 0.07 mm of upward keeper deflection at the ears at the maximum-compression load (with a PLA modulus of about 3000 MPa assumed). That is within the 0.25 mm free play between the ears and the keeper; with only the four end screws it would be about 1.5–2 mm. The keeper side members are 3.0 mm thick. A keeper with 5.0 mm side members fits the same screws if more stiffness is wanted: its counterbores are deeper, so the head seat height is unchanged.

The screw heads sit in an approximately 0.5 mm-deep recess in the keeper, and the captive hex pockets are approximately 2.5 mm deep. This leaves approximately 1.5 mm of PLA above each nut pocket and gives nearly full use of the 12 mm screw length, with only slight nominal screw projection beyond the nut. The keeper seats on broad printed lands, so the fasteners clamp it against the carriage rather than suspending it between isolated points.

The vertical guide stack is: 4.0 mm base plate; 2.0 mm rise from the plate front to the lower guide-ear running surface; 3.0 mm plunger guide ears in a 3.4 mm guide slot; and a 3.0 mm keeper frame.

Use a very light film of plastic-safe lubricant on the PLA-on-PLA sliding interfaces, particularly the plunger ears/tabs in their guide slots and any other plunger sliding faces. The lubricant is McMaster 1418K51 clear silicone grease, NLGI Grade 1. Smooth/dress the printed sliding surfaces first, apply only a trace amount, cycle the mechanism, and wipe away visible excess so the lubricant acts as a thin boundary film rather than a grease-packed slide. Keep lubricant away from CA/epoxy/RTV bonding surfaces and apply it only after nearby adhesive work is complete.

The plunger seating lips and the cassette edge bevels are 27.5° from the cassette plane, with a 1.25 mm bevel depth.

For keeper-frame sizing, McMaster lists the 2006N292 spring rate as 0.14 lbf/mm. At the 9 mm fiducial compression this corresponds to approximately 5.6 N of in-plane plunger force; at the 14 mm most-compressed end of normal travel it is approximately 8.7 N. A frictionless first-order wedge model at 25–30° from the cassette plane gives approximately 9.7–12.0 N upward reaction per plunger at fiducial and approximately 15.1–18.7 N per plunger at maximum normal compression. These reactions are carried upward through the plunger ears into the keeper frame; about 20 N per plunger is the working-load scale before safety margin.

8.5 Spring seating and retraction

The spring plunger is preloaded by the McMaster 2006N292 compression spring (§8.3), acting between the frame's spring end wall and the plunger's outboard face on the adjuster axis, 8.77 mm above the plate back.

Each spring end sits in a printed cup, 3 mm deep, sized to the measured 0.272 in (6.91 mm) coil OD plus 1.0 mm diametral clearance:

- one cup extends inboard from the end wall: its side walls are on the guide frame, and its roof is carried by the keeper's spring-end member, so the keeper closes over the spring end when installed. The cup's boss (11.5 mm wide, roof at 13.9 mm) rises 1.5 mm above the end member and continues back over its full depth, so the coil's end face bears on a wall as tall as the cup; 0.6 mm fillets blend its sides into the end member top;
- one cup extends outboard from the spring plunger;
- both cups are open toward the plate deck, which carries the coil. Their side walls and roofs locate the coil ends laterally and against escape toward the cassette-loading side. Both print with legs on the bed and the roof bridging the coil, without supports.

The cup roof is 1.2 mm over the coil and the side walls 1.8 mm, so the feet keep a 1.0 mm flat inside the 0.4 mm underside chamfer (§8.4).

The spring has a free length of approximately 25.5 mm. The carriage is designed around a 16.5 mm spring length at the fiducial / mid-travel cassette position, corresponding to approximately 9 mm spring compression from free length.

The spring's free length is about 4–4.6 times its mean coil diameter, and both ends sit on parallel faces because the plunger is guided. That is below the roughly 5.3 critical ratio for buckling with fixed ends, so the cups need only locate the ends rather than guide the full length. The spring itself does not guide the plunger; the carriage's guide slots constrain plunger translation and rotation.

Normal supported plunger travel is ±5 mm about the fiducial position, for 10 mm total working travel. The corresponding spring-length range is approximately:

- 21.5 mm at the least-compressed end of normal travel;
- 16.5 mm at fiducial / mid travel;
- 11.5 mm at the most-compressed end of normal travel.

Loading retraction is 6 mm from fiducial, 1 mm beyond normal travel (10.5 mm spring length), leaving 4.5 mm between the cups. Any additional compression capacity is treated as design margin rather than normal commanded travel. The spring plunger's ears reach the end of their slot before the spring is compressed solid, so the slot end stop, not the spring, limits retraction.

Retraction for cassette loading uses a thumb tab on the spring plunger's front face at its outboard end:

- 20 mm wide × 3 mm thick × 6 mm tall, centered between the guide ears and outside the cassette footprint;
- the thumb reaches over the cassette and pushes the tab's inner face outboard, at about 9 N at full retraction;
- 0.5 mm chamfers on the exposed edges and 0.5 mm root fillets on its thumb-side face and ends; the spring-facing face is left square;
- a Ø1.6 mm grip bead along the inner top edge, flush with the tab top, prevents finger slip.

The tab is kept low, close to the plane of the guide ears, so that pushing on it does not twist the plunger and jam the ears.

8.6 Common cassette

Cassette envelope:

- 64 × 64 × 5 mm PLA
- 24 mm diameter central clear aperture
- identical shallow bevels on all four perimeter edges
- insertable at 0 / 90 / 180 / 270°
- perimeter kept clear enough to use any opposing edge pair
- orientation/polarity indicated by markings rather than a mechanical key.

Cassettes load with the spring plunger retracted 6 mm beyond fiducial (§8.5).

Cassette variants, all built on the common blank:

1. single razor knife edge;
2. centered fine-wire or filament dark-field cutoff (§8.8);
3. striated / multicolor gel filter (§8.9);
4. dark-center color filter (§8.9).

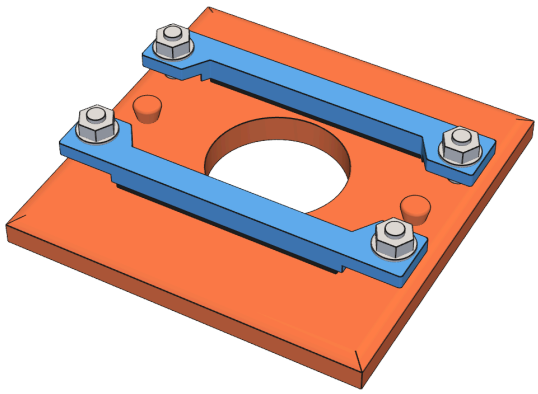


Fig. 13. Cassette blank with its two clamp bars and hardware

The common cassette blank carries one opposed pair of integral filament cleats on the front / non-datum face, rather than making cleats a special cassette variant: two cleats, one on each side of the 24 mm optical aperture, at (±25, 0) mm on an aperture centerline, beyond the ends of the razor blades and between the clamp bars. The cleats are slightly conical, Ø3.5 mm at the cassette face widening to Ø5 mm toward the top and rising 3 mm, so filament tension drives fine wire or thread down against the cassette front face. The tops are rounded to avoid damaging fine wire. Route a filament in an S path, around opposite sides of the two cleats; the free span then

crosses approximately through the aperture centerline. Small angular error from finite cleat diameter or wrapping is acceptable because the SM1RC/M rotates the complete carriage. This makes the same common cassette blank usable for blade, filter, Photofoil, wire, and thread experiments while keeping the rear datum face unobstructed.

8.7 Cassette clamp bars

These bars and their hardware are the clamping standard for the cassettes (§8.6); the slit head uses printed toe bars (§7.4).

Common clamp bars:

- two identical printed rigid PLA bars fit each cassette;
- each bar uses two fasteners, one toward each end, so clamp force is distributed along the element;
- no holes are made through razor blades, filter material, or other optical elements;
- no blade-specific locating pocket or guide is required;
- McMaster 9852N37 adhesive-backed 1/32 in, 60A solid EPDM is applied to the removable clamp-bar face to provide compliance and friction;
- the bar is meant to bear on the broad flat portion of a razor blade rather than the fold: **Provisional**, because the blade's fold is 7.04 mm deep, so a knife edge on the aperture axis has the fold under the bar, and the bar's height above the cassette face is not yet set from the blade;
- the same clamp-bar geometry holds razor knife edges, Roscolux gel/filter material, Photofoil, and similar thin sheet cutoff elements.

Clamp-bar fastening is from the cassette rear/datum face toward the optical/front face so no screw ends or nuts can protrude from the sliding/datum surface. Fasteners are positioned outside the optical-element footprint where practical; the clamp bar spans and pinches the element against the cassette front face. At each fastener, use:

flush M3 countersunk flat-head screw → 5 mm cassette body → printed clamp bar → M3 flat washer → ordinary M3 hex nut

The M3 flat-head screw is seated fully flush or very slightly below the rear cassette face and retained in its countersink with a small CA tack using McMaster 1818A45 / Loctite 4061. The CA serves only to keep the screw captive and resist rotation while the front-side nut is adjusted; it is not relied upon as the structural clamp load path. The ordinary M3 nut remains exposed and accessible on the front of the clamp bar. Place the washer between the nut and printed bar. Do not use prevailing-torque/nyloc nuts or threadlocker here.

Cassette clamp-bar hardware

Component	McMaster part	Specification
Flat-head screw	92125A133	M3 × 0.5 × 14 mm, fully threaded 18-8 stainless, 90° countersunk head; 6 mm head diameter, 1.7 mm head height, 2 mm hex drive
Flat washer	93475A210	18-8 stainless, 3.2 mm ID × 7 mm OD × 0.4–0.6 mm thick; DIN 125 / ISO 7089
Ordinary full-height hex nut	91828A211	M3 × 0.5, 18-8 stainless, 5.5 mm across flats × 2.4 mm high; DIN 934 (the common split-clamp nut, §5.2)

The clamp bars are 4 mm printed PLA. The cassette blank has four Ø3.4 mm screw holes at (±27, ±12) mm with rear-entry 90° countersinks matched to the 92125A133 head (Ø6.2 mm at the face: 6 mm head plus 0.2 mm diametral allowance), and each bar end provides a washer bearing land at least 7.2 mm across. The screw head finishes flush with or slightly below the rear/datum face so it cannot contact the carriage datum pins or interfere

with cassette sliding. The fasteners, bars, nuts, cleats, and projecting screw ends are clear of the 24 mm optical aperture and carriage interfaces. The holes and countersinks are features of the common cassette blank, not blade-specific geometry.

Because the clamp fasteners do not pass through the optical element, the screw length is not sized from a blade-thickness stack. The 14 mm length gives full nut engagement and holds razor blades under the 4 mm bars, with the head seated flush in its countersink. The screws are perpendicular to the cassette face and the lips act at the cassette edges, so the screw length does not bear on the lips; the bar ends, which reach the bevel zone, are relieved to clear them.

8.8 Fine wire

The dark-field filament set is enameled copper magnet wire in three readily available AWG sizes. The nominal diameters below are bare-conductor diameters; enamel makes the finished optical obstruction slightly larger.

Wire size	Nominal bare diameter	Approx. metric diameter	Experimental role
44 AWG magnet wire	0.0020 in	0.0508 mm	fine / high-sensitivity filament
34 AWG magnet wire	0.0063 in	0.160 mm	intermediate filament, near the expected slit-image width
30 AWG magnet wire	0.0100 in	0.254 mm	coarse filament for stronger gradients / deeper dark field

Exact finished outside diameter is not critical for the qualitative instrument.

The common cassette blank's cleat pair, described in §8.6 is the quick-change mounting for these wires and for dark sewing thread or other fibers. Apply only modest tension, particularly to 44 AWG wire; the conical cleats seat the filament against the cassette face, and the S-wrap places the free span approximately on the aperture centerline.

8.9 Color-filter material

Filter material is cut from the Roscolux 3 × 6 in Designer Color Selector, which provides more useful material per color than very small swatches.

For color schlieren:

- use the white Nichia LED;
- lock camera white balance;
- lock exposure.

9. Camera and telephoto support

The camera and lens are described in §9.1–9.2. Their support is a lens cradle and a phone rest (§9.4–9.8), with an alignment procedure (§9.9). **Provisional:** the alignment procedure (§11).

9.1 Camera

- Apple iPhone 17, base model, in an OtterBox Commuter case
- held in landscape, with the rear cameras toward the mirror and the camera end of the phone toward the optical axis
- the Main camera is used, at 1× (or its 2× crop). It is the lower of the two rear apertures with the phone in portrait, "rear camera 2" on Apple's drawing (`docs/reference/Apple-iPhone-17.pdf`); the upper aperture is the Ultra Wide, used only at 0.5×
- measured camera optical axis approximately 60.6 mm above the lower long edge, in the case
- measured Main camera axis approximately 1.350 in / 34.3 mm from the near short edge, in the case; the case shape makes this hard to measure accurately with calipers
- measured case depth, flat back to front, 0.540 in / 13.72 mm (bare phone 7.95 mm), and back plate thickness 0.125 in / 3.18 mm; the case corners are rounded
- measured mass 188.7 g bare, with its glass screen protector, and 238.6 g in the case

Manufacturer dimensions of the bare phone, from the drawing:

- both rear cameras are 13.62 mm from the nearer long edge, on a line parallel to it, so the measured axis height applies to either aperture
- the Main camera is 31.34 mm from the top short edge, 17.72 mm beyond the Ultra Wide at 13.62 mm
- body 149.61 × 71.45 × 7.95 mm; the camera glass stands 3.45 mm above the back glass

The drawing and the two measurements agree. The cameras are 57.83 mm from the farther long edge of the bare phone, so the measured 60.6 mm implies a case wall of about 2.8 mm at that edge; the 31.34 mm and the measured 34.3 mm imply about 3.0 mm at the short edge (both calculated).

The phone is modeled in `src/schlieren/parts/phone.py`, whose comments describe the bare phone, the case, and the camera cover in detail. The case's measured outside is 155.5 × 77.0 mm. A clear adhesive polycarbonate cover supplied with the case sits on the camera plateau. The lens reaches through the opening in the case back and seats on the cover top, which is the back plane of the cased phone, and the cased phone is 14.35 mm thick behind it. The lens seats on the cover's measured outer face, so the cover's inner details, such as its counterbores, do not affect the spacing.

Initial recording modes:

- 4K/30 for normal work;
- 4K/60 for faster flow phenomena;
- SDR / HDR off as a stable starting point.

9.2 Telephoto

- iOgrapher proSnap 7×
- 17 mm camera-standard mount
- approximately 37.0 mm barrel diameter
- measured mass approximately 215 g

Measured sections, from the aft end forward:

Section	Length	Outside diameter
threaded boss ring (17 mm)	0.10 in / 2.5 mm	17 mm thread
shoulder	0.40 in / 10.2 mm	1.26 in / 32.0 mm
main barrel	4.285 in / 108.8 mm	about 37.0 mm

The overall length is 4.785 in / 121.5 mm, the sum of the sections; a direct end-to-end measurement agrees within 0.3 mm. The boss threads into the lens mount on the phone case, and the shoulder seats against the mount; the model takes the aft end of the boss to be on the camera cover, at the back plane of the cased phone. The main barrel is 1.457 in / 37.01 mm in diameter at its rear end and 1.459 in / 37.06 mm further forward. The focus ring, 0.455 in / 11.6 mm long, ends 0.175 in / 4.4 mm behind the front of the barrel, so it starts 105.5 mm from the aft end. The barrel is clampable from 12.7 mm to the focus ring, about 93 mm, and the focus ring must stay clear. The front glass is a clear aperture of 0.845 in / 21.5 mm at the bottom of a stepped recess. The focus ring moves the glass, and at the working focus of about 3.2 m the ring is fully forward and the glass is 0.5 in / 12.7 mm behind the front of the barrel, 108.8 mm from the aft end.

Optically the lens is a Galilean telescope of 7× angular magnification: the image through it is upright and the exit pupil is virtual (observed). The front glass is the aperture stop, so the exit beam is about 3.1 mm in diameter (calculated, 21.5 mm / 7), and the exit pupil lies inside the barrel, in front of the rear glass. There is no field stop; the rear glass limits the field, and off-axis rays are clipped gradually. The rear glass has a measured clear aperture of 0.471 in / 11.96 mm. With the lens seated in its clamp, the center of the rear glass is 0.125 in / 3.18 mm from the camera cover.

The lens is modeled in `src/schlieren/parts/lens.py`, for visualization only.

The lens barrel, rather than the phone shell, is the preferred transverse optical datum because accurate lens centering minimizes vignetting.

9.3 Alignment requirements

The returned slit image at the cutoff is the aperture stop for the camera: every ray from the test region passes through it, then spreads again at the ±1.82° cone angle (§3.2). The values below are calculated. **Provisional:** they rest on assumed values, namely a barrel front about 42 mm behind the cutoff plane (a 10 mm gap to the cutoff split ring's rear face, which lies about 32 mm behind the cassette front face; the front glass is then about 55 mm behind the plane), an 11 mm bright slit image (the 10.4 mm lit length of §6.3, enlarged about 5 % by the field lens, §7.9; its shoulder reaches about 16.5 mm, which would cut the along-slit centering margin to about ±0.75 mm for the shoulder's full extent), and a Main-camera pupil of about 3.7 mm (6 mm focal length at f/1.6). The telephoto's front clear aperture is measured at 0.845 in / 21.5 mm. The distance is not critical: the cone footprint grows by only about 0.06 mm per mm of distance, so a 10 mm shorter gap would add about 0.3 mm to the along-slit centering margin.

Quantity	Value
Beam cone at the front glass	about 3.5 mm diameter
Beam footprint at the front glass	about 3.5 × 14.5 mm
Slit image at the phone, reduced 7× by the lens	about 1.6 mm long
Frame at 1×, 4K 16:9, through the lens	about 9.9° × 6.1°
Mirror in the frame	3.6° of the 6.1° height

The resulting tolerances on the camera and lens as one body:

- centering on the beam: about ±3.5 mm along the slit direction and ±7 mm across it, before the beam footprint reaches the edge of the front aperture (calculated from the 21.5 mm clear aperture). The phone pupil allows more, about ±7.5 mm and ±13 mm, before the reduced slit image leaves it;
- pitch: about ±1.2°, the spare frame height either side of the mirror;
- yaw: about ±3°.

Pointing is therefore the governing adjustment, and centering is moderately forgiving. Image quality still favors centering well inside the vignetting limit.

The rear glass (§9.2) also limits the field. A ray at image-space angle ϕ (about 7× the object-space angle) crosses the rear glass about D tan ϕ from the axis, where D is the distance from the rear glass to the Main camera's entrance pupil, and the bundle through the pupil is clipped once D tan ϕ plus its footprint radius of about 1.9 mm reaches the 6 mm glass radius (calculated). The mirror region (±1.8°) stays clear up to D of about 18 mm, the frame half-height (±3.05°) up to about 10 mm, and the frame half-width (±4.95°) up to about 6 mm, beyond which the left and right edges and corners darken. **Provisional:** D is the 3.18 mm seat offset plus the depth of the camera's entrance pupil behind the cover's outer face, which is not yet known; about 6–8 mm is assumed. A flat-field photograph through the lens would give D from the position of the vignetting edge. At D of 8 mm, the mirror edge leaves about 2 mm of decentering between the phone and the lens axis.

9.4 Support

The phone and the lens threaded onto its case are one rigid body. The lens barrel is its datum (§9.2): the lens is held at two stations along its barrel, and the phone hangs from its lens mount. Two separate rail fixtures carry them:

- the lens cradle, specific to the lens, holds the barrel at both stations on four adjusting screws (§9.5), with two retaining bands holding the collars down on them (§9.8);
- the phone rest, specific to the phone, supports the phone's lower edge at one point (§9.6). A different phone needs only this part redesigned.

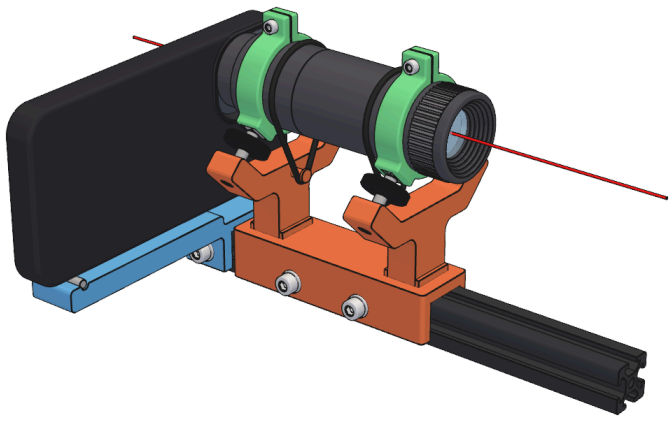


Fig. 14. Lens cradle and phone rest on the imaging rail, with the phone and the lens

The body is located by six contacts and no more, so adjusting one station does not strain the lens mount:

- four screw tips, two at each station, fix the lens axis in x and z at both stations, and so its position, pitch, and yaw;
- one of those tips sits in a groove, fixing the lens fore/aft;
- the phone rest fixes roll.

Each station's pair of screws places one point of the lens axis, so the two stations give position and pointing without coupling between them beyond the lever ratio. A second groove would fix the fore/aft position a second time against the spacing of the collars on the lens, and would stop the pad sliding that yaw adjustment needs. Holding the lens, not the phone, takes the lens weight off the lens mount: the mount carries about 1.3 N in shear (calculated), against roughly 170 N·mm of bending if the lens were cantilevered from the phone.

The model is `src/schlieren/parts/camera_support.py`. It holds the printed parts, with their fits, pockets, fillets, and wall thicknesses, and checks contacts, clearances, and the lens and band loads; purchased hardware is envelopes or vendor models.

9.5 Lens cradle

The cradle is five printed parts and two bands. Two split collars clamp the barrel; two yokes carry the adjusting screws; one shoe straddles the rail and carries both yokes.

Collars

Each collar is a thin ring on the barrel with a clamp ear on top and a flat pad platform at each screw:

- the aft collar centered about 24 mm from the lens aft end, which leaves finger room for its thumb nuts ahead of the phone; the fore collar centered about 96.5 mm from the aft end, with its front face about 3 mm behind the focus ring; about 72.5 mm between them;
- 12 mm wide, with a 3 mm wall around a bore of 37.3 mm, the measured 37.0 mm barrel plus a 0.3 mm diametral allowance, so a collar is 43.3 mm across. The wall is thin because the pad platforms carry the local thickness; the pad loads are a few newtons, so stiffness is not limiting;
- the ear takes the common M3 split-clamp hardware (§5.2) with the rail shoe's split and ear thicknesses: the screw passes through the thin ear, and the hex nut is captured in a pocket in the thicker ear across the split, open to the outside, with a roof of at least 1.5 mm over it. The screw axis stands 4.25 mm above the ring top, so the screw head clears the ear's root fillet. The printed collar has the thin ear on its left, seen from the aft end looking forward; each collar is installed turned half a turn about the vertical axis, which puts the screw ear and head on the outboard (right) side, facing away from the source rail for the hex key;
- inside corners, the ear and pad platform roots, are filleted at 2 mm for stress control; the outside corners and the outer edges of both ring ends at 1 mm, for the fingers. The faces of the split stay sharp so that the gap stays fixed.

The two collars are the same part except that the aft collar's printed left pad is a rod seat in place of a flat bearing insert (below). Installed turned half a turn, it is the aft collar's right, outboard pad.

Pads and pockets

Each screw tip bears on a flat pad on the collar, tangent to the ring at the screw: the pad face stands one bearing-insert thickness outside the ring, and the pocket floor is tangent to the ring's outside. Each pad sits in a pocket with a rim around it, so a ball end that wanders reaches a wall instead of the edge of the pad:

- three pads are flat bearing inserts, the 10 × 5 × 2 mm N52 bar magnets (§8.3), set with the 10 mm side across the rail, bonded with low-viscosity CA wicked in, into pockets 0.2 mm larger across each dimension (0.1 mm per side, the allowance every printed bearing-insert pocket in the design uses);
- the fourth, at the aft right (outboard) screw, is a groove between two dowel pins of 3 mm diameter and 10 mm length, lying across the rail on the pocket floor. The ball sits between them, free to slide across the rail but not along it. A printed rib 1.0 mm high and about 1.7 mm wide runs between the pins along the pocket, and each pin lies against its top corner: the rib sets the pins' spacing (4.5 mm between axes) and keeps them parallel, so bonding them needs no spacer. The ball stays 1.0 mm clear of the rib;
- the rim stands 0.8 mm above the pad face. The ball-tip screw's end face is 1.2 mm behind the ball apex, so the rim stays below it;
- above the pad face the pocket opens outward by 0.23 mm per side, so that the rim edge stops the ball apex 1 mm short of the pad edge, the margin the travel limits below assume. The walls around the pockets are 1.5 mm.

The two screws of a station are at right angles, so advancing one slides the collar across the other's pad by the same distance. Pad size, not screw travel, limits the adjustment. Calculated, keeping each ball 1 mm inside the pad edges, which the rim enforces:

Adjustment	Range
Each screw	about ±3 mm
A station, in x or in z alone	about ±4 mm
Lens tilt, pitch or yaw	about ±2°
One turn of both screws of a station	about 1.1 mm, or about 0.9°

The tilt limit comes from the 5 mm side of the pads, on the pads that slide along the rail as the lens yaws about the groove. These ranges cover the alignment tolerances (§9.3) and the expected build errors.

Yokes

Each yoke is one printed part, 14 mm thick along the rail, in the shape of a T with an inclined arm either side, printed on its side so that every layer lies in the plane of the loads:

- a column 20 mm wide, the width of the rail, rising from the shoe to a crossbar 10 mm thick whose top stands 13 mm below the bottom of the collar, clear of the thumb nuts;
- each arm is a slab 10.5 mm thick, square to its screw. Its lens-side face carries the insert and runs down to the crossbar top; its back face runs down to the crossbar underside, so there is no stub or step;
- fillets of 5 mm at the column root, 4 mm where the arm blends into the crossbar, and 2 mm around the outer arm corners;
- each arm holds a heat-set insert (§9.7) in a 6.4 mm hole from the lens-side face with the flange recessed flush. The wall around each insert and its flange recess is at least 2.9 mm, and a 5.5 mm bore carries on through the back of the arm for the screw and its hex key.

Under the adjuster reactions of about 8 N the crossbar root sees about 0.6 MPa and the arm tip moves about 0.01 mm (calculated, with a PLA modulus of about 3000 MPa assumed), so stiffness and strength are not limiting; the fillets are margin.

Shoe and joint

The shoe's lower section is that of the common rail shoe (§5.2): a 20.4 mm rail opening, 5 mm side walls, 18 mm skirts, a 5 mm deck, and 2 mm rounded outside corners. M5 clearance holes cross the skirts at rail mid-height for the side-channel screws, two of them 36 mm apart midway between the collars. The shoe extends 9 mm beyond each collar face, so it is symmetric about the middle of the two stations. Each yoke is bolted to the deck, so that it prints on its side and the shoe prints without supports:

- one 90° countersunk M5 screw from the underside of the deck, on the rail centerline, with its head flush with the underside, into a heat-set insert in the column foot, with the insert's flange bearing on the deck;
- two straight walls on the deck across the rail, ahead of and behind the column, 1.6 mm thick and 3 mm high, with 0.2 mm clearance to the column faces. They locate the yoke along the rail and stop it twisting;
- the screw ends inside the insert with about 5 mm of engagement.

The fitting and removal sequence is set by the bands (§9.8).

9.6 Phone rest

The phone hangs from the lens with its body to the outboard side of the optical axis (§3.4). Its weight, centered about 43 mm from the lens axis, would turn the lens in its collars. The phone rest stops that: the phone's lower long edge sits on one dowel pin lying along the rail, about 100 mm outboard of the axis, on the arm of a short printed shoe of its own. The shoe's lower section is that of the cradle shoe (§9.5), with a single pair of M5 clearance holes for the side-channel screw.

- The shoe is 22 mm long and centered under the rod. The arm runs across the rail from the shoe's outboard wall to 8 mm beyond the rod, 10 mm deep and as long as the shoe. The rod lies half sunk in a half-round seat in its top, with 0.1 mm radial clearance, and is bonded with low-viscosity CA wicked in.
- The part is printed on its side, like the yokes: it is one profile in the plane of the load, extruded along the rail, so every layer is that profile and the arm needs no bridge or support. Only the side-channel screw bores are horizontal. The shoe's vertical corners on the arm side are left sharp, where the arm joins the wall.
- The arm's inside corners are filleted against stress: 3 mm under the arm where it meets the shoe wall, the largest because it takes the bending, and 2 mm where the arm top steps up from the deck. The outside corners at the tip and the step are rounded 2 mm, and the seat edges 0.5 mm. The arm is kept 10 mm deep so that the root fillet stays at least 1 mm above the side-channel screw head on the outside of the skirt.
- With the measured 60.6 mm camera-axis height, the phone's lower edge is about 11.75 mm above the rail top.
- The rest carries about 1.0 N (calculated, same assumptions as the contact loads in §9.8). It matters to the cradle, because it keeps the fore pad load from falling toward zero.
- The rest is fixed. Raising or lowering the aft collar rolls the phone slightly about the lens axis, which does not matter; the phone is not gripped and slides on the rod.
- The rest holds no optical alignment, so its dimensions are not critical.
- Because the rod lies along the rail, the rest needs to be only roughly under the phone.

9.7 Adjuster hardware

There are four adjusting screws, two under each collar. Each is a ball-tip set screw in a heat-set insert in its yoke arm, with a thumb nut threadlocked mid-screw as a handwheel.

Hardware:

- McMaster 93339A252, M5 × 0.8 × 25 mm ball-tip set screws: alloy steel, 25.0 mm overall including the ball, 2.5 mm hex socket, and a 3.0 mm ball (ball size from the vendor model, `cad/vendor/McMaster-93339A252.step`; drawing `docs/reference/McMaster-93339A252.gif`)
- McMaster 94459A797, M5 flanged heat-set inserts: 6.73 mm long overall, with a 7.92 mm flange 1.02 mm thick, a 7.11 mm knurled body, and a 6.32 mm pilot (drawing `docs/reference/McMaster-94459A797.gif`; vendor model `cad/vendor/McMaster-94459A797.step`). The drawing calls for a 6.40 mm hole and at least 6.48 mm of material. Each is set from the lens-side face of its arm, flange outward and recessed, so the screw's reaction presses it into its hole. Two more are set in the yoke feet for the joint screws.
- McMaster 92125A208, M5 × 0.8 × 10 mm 18-8 stainless hex-drive flat-head screws (90°, 10 mm head), two for the yoke joints (§9.5)
- McMaster 92815A202, M5 thumb nuts used as threadlocked mid-screw handwheels
- Loctite 271, McMaster 91458A160
- McMaster 8915N111 heat-set insert installation tip set (M5 tip), on a Hakko iron
- Amazon B0DMCY4FN1, N52 nickel-plated bar magnets, 10 × 5 × 2 mm, as the flat bearing inserts, one at each of three screws
- McMaster 91585A351, Ø3 × 10 mm 18-8 stainless dowel pins (ISO 2338 m6, chamfered ends), two for the groove at the fourth screw
- McMaster 91585A457, Ø4 × 20 mm 18-8 stainless dowel pin, for the phone-rest rod
- the common M3 split-clamp screw and nut (§5.2), one pair per collar

9.8 Retaining bands

The collars sit on their pads under gravity alone, so each pad pair is held down by an elastic band. The pairs carry only the lens and phone weight, and the fore pair least of all: contact loads, calculated from the measured 215 g lens and 238.6 g cased phone, with the lens balance point measured at about 60 mm behind its front (about 61.5 mm from the aft end), are

- aft screw pair: about 2.9 N;
- fore screw pair: about 0.5 N. The phone hangs behind the aft collar, so its weight pivots the lens about the aft screws and unloads the fore pair, but it cannot lift it: the lens moment is the larger.

A downward push on the phone, such as a screen tap, lifts the fore end when it exceeds roughly 2.5 times the fore pair's load, about 1.3 N unaided, and a sideways nudge unseats a pad when it exceeds the load on its pair. The alloy-steel screw tips are attracted to the magnetic bearing inserts, which adds some preload; it is not relied upon until measured.

Each band is an endless O-ring, 2 mm in section, of oil-resistant 70A Buna-N with an inside diameter of about 40 mm, lying around the bare barrel beside its yoke on the side toward the other collar, 11.8 mm from the collar's center. Its two legs run down inboard of the thumb nuts to a peg on the center of the yoke's crossbar face. The peg has a 4 mm stem and a 6 mm lip; the band lies on the stem under the lip.

- The band acts on the barrel, which the collar's clamp carries to the pads, exactly as it carries the lens weight. The band needs no groove on the collar and settles into the plane through the peg.
- The legs are about 65° above horizontal, so 5 N of hold-down takes about 2.75 N in each leg. A band of that size stretched about 19% pulls about 3 N (calculated, with a 70A rubber modulus of about 5 MPa assumed).
- With one band beside each yoke, between the collars, each pad pair receives the hold-down of 5 N in total from the two bands (calculated), about 4.2 N from its own band and about 0.8 N from the other. That raises the tap threshold at the fore end to roughly 14 N.
- The legs clear the thumb nuts by more than 5 mm, so the band stays out of the way of the fingers.
- To fit, slide the bands over the front of the lens onto the bare barrel before clamping the collars on; refitting one afterwards means stretching it around the other collar and its ear. A band stays on the lens when the lens is lifted out with the peg end unhooked.

9.9 Lens alignment

This is the alignment the cradle must make possible, kept here as a feasibility check on the adjusters. **Provisional:** the procedure is untested, and its targets assume that the printed shoe seats fully on the rail top and the collars seat on their barrel sections.

The tolerances (§9.3) are loose compared with the adjuster resolution, so the alignment is mechanical first and then finished through the camera, with no laser.

Each adjuster moves the lens axis along its own screw, so at one station, with the screws at ±45° and a thread pitch of 0.8 mm (calculated):

- both screws turned the same way, n turns: z moves n × 1.13 mm;
- the two turned in opposite directions, n turns each: x moves n × 1.13 mm;
- one screw alone, one turn: x and z each move about 0.57 mm;
- pitch and yaw come from setting the two stations 72.5 mm apart differently; 1 mm of difference is about 0.8°.

The sequence:

1. Fit both collars and the phone on the lens at their stations, set the four thumb nuts to equal exposed screw length (mid-travel), and note their positions. Put the cradle shoe on the rail with its clamp screws loose.
2. Slide the shoe along the rail to put the lens front at the 10 mm gap assumed in §9.3 from the split ring's rear face, then snug the clamp screws. The groove at the aft right screw fixes the lens fore/aft on the cradle, so nothing later moves it.
3. Center the barrel at each collar mechanically, aft first, and re-measure the other collar after each change. Height: from the shoe deck, beside the yoke, up to the barrel top, whose target is 72.35 + 18.5 − 5 = about 85.85 mm (the deck top is 5 mm above the rail top; the barrel diameters of 37.01 and 37.06 mm differ by too little to matter). Lateral: with the shoe pressed to one side of the rail, taking up its 0.4 mm lateral play, from the shoe's outer wall to the barrel side, whose target is 18.5 − 15.2 = about 3.3 mm. A ±0.5 mm error between the stations is about 0.4° of pitch or yaw, within tolerance.
4. Finish through the camera, because the phone camera may not lie on the barrel axis in its mount and case. With the source lit and the cutoff cleared, view the returned field at 1× and trim the aft screws in x and z until the vignetting at the corners is even; a turn of one screw is only about 0.6 mm, so turn in small steps. The mirror should sit centered in the frame, which sets pitch and yaw to within the spare 1.2°.
5. Check the roll: the phone-rest rod should just touch the phone's lower edge without lifting it, a paper strip sliding between them with light drag. A lower-edge height different from the model is taken up by shimming the rod seat.
6. Record the final thumb-nut positions, so that refitting the lens needs only a check.

10. Mirror cell

The mirror cell is fabricated from Baltic-birch plywood panels, and uses a three-point fine-adjustment architecture with compression springs and spherical washer interfaces. The model is `src/schlieren/parts/mirror_cell.py`.

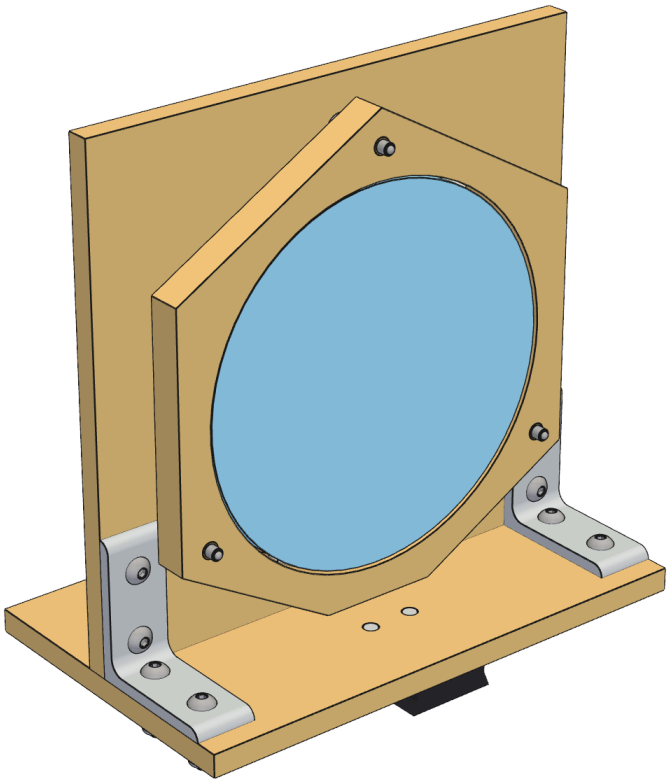


Fig. 15. Mirror cell from the mirror side: base plate, stationary plate, mirror plate and mirror, and angle brackets

10.1 Plates and basic architecture

The mirror cell is constructed from three Baltic-birch plywood plates:

- base plate: 11.5 in wide × 7.0 in front-to-back × 1/2 in;
- stationary plate: 11.5 in wide × 11.0 in high × 1/2 in;
- mirror plate: 3/4 in cut to the truncated-hex geometry below.

The base plate provides the structural/tripod-support base and is joined to the stationary plate by angle brackets described in §10.7. The stationary plate carries three fine-adjustment stations (§10.2) which couple it to the mirror plate. The 3/4 in mirror plate carries the 203 mm mirror and moves relative to the stationary plate.

The stationary plate is centered left/right on the 11.5 in-wide base. The rear face of the 1/2 in stationary plate is 4.000 in aft of the front edge of the 7.0 in-deep base, leaving 3.0 in of base behind it. Two McMaster 8681N11 angle brackets face forward, toward the mirror plate; their measured forward footprint is approximately 3.5 in from the stationary-plate rear face, leaving approximately 0.5 in of base margin at the front edge. This orientation places the brackets on the mirror side of the stationary plate so they directly resist the forward overturning moment of the mirror assembly.

Using measured masses of 1.56 kg for the approximately 18 mm-thick mirror, 840 g for the assembled stationary and mirror plates and adjuster hardware, about 318 g for the 11.5 × 7.0 × 1/2 in base plate, and 79 g each for the two aluminum brackets, the estimated completed mirror-cell fore/aft center of gravity is approximately 3.0 in / 76 mm aft of the front edge of the base. A working uncertainty of approximately ±0.2 in / ±5 mm is conservative. This places the expected center of gravity approximately 1.0 in / 25 mm forward of the rear face of the stationary plate.

The cell uses:

- three fine-adjustment stations arranged at 120° intervals;
- compression-spring preload at each station;
- spherical washer interfaces for angular accommodation and gravity-load transfer;
- tripod interface through the CAMVATE 577/501/504-compatible quick-release system and the Sunwayfoto LB-68R leveling base.

The mirror-plate geometry is:

- start from a regular hexagon 11.000 in / 279.4 mm vertex-to-vertex, oriented with one vertex upward;
- the corresponding normal flat radius / apothem is 4.7631 in / 120.98 mm;
- three alternating vertices lie on the three 120° adjuster axes, with the top vertex occupied by one adjuster axis and the two lower-side occupied vertices by the other two;
- the three adjuster-hole centers lie on a 4.750 in / 120.65 mm radius from the mirror center;
- the mirror aperture is 8-1/8 in / 206.38 mm diameter, providing approximately 1/16 in radial clearance around the nominal 203 mm mirror;
- clip the unused bottom hex vertex with a horizontal chord whose midpoint is 4.7631 in / 120.98 mm below the mirror center, matching the hexagon's apothem. This removes approximately 0.7369 in / 18.72 mm of the lower vertex and produces a lower edge approximately 2.553 in / 64.8 mm long;
- the resulting trimmed mirror-plate height is approximately 10.263 in / 260.7 mm from the truncated lower edge to the top vertex.

The nominal clearance from the bottom trimmed edge of the mirror plate to the top surface of the base plate is 3/8 in / 9.53 mm. This places the mirror center approximately 5.138 in / 130.51 mm above the base top. The top mirror-plate vertex then lies approximately 10.638 in / 270.21 mm above the base top, leaving approximately 0.362 in / 9.2 mm of stationary-plate height above it within the 11.0 in-high stationary plate.

This truncated-hex geometry preserves the same approximately 0.701 in / 17.8 mm radial plywood width between the 8-1/8 in mirror aperture and the truncated bottom edge as exists between the aperture and each hexagon flat. It therefore improves base clearance without introducing a uniquely narrow section.

10.2 Three adjuster stations

There are three equivalent adjuster stations.

The adjusters are Kozak parts:

- TS250-80-2500 1/4 in-80 adjuster
- TB250-80-625 bushing
- KB250-80 adjustment knob

Use one adjuster, one bushing, and one knob at each of the three stations.

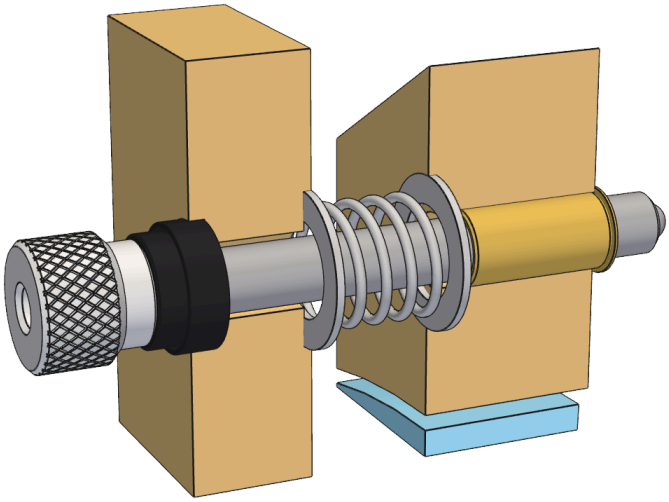


Fig. 16. Adjuster station, plywood and mirror cut away: knob, spherical washer pair, spring and seat washers, bushing, and adjuster screw

Each adjuster station includes a concentric compression spring.

The adjuster bolt passes through the spring and therefore positively retains and laterally locates it. There is no separate spring pocket or spring bore in the plywood.

Adjuster-bushing mounting

Each Kozak TB250-80-625 bushing is bonded directly into the 3/4 in mirror plate.

Mounting details:

- the bushing's narrow flange is on the mirror-facing / outboard face of the mirror plate, opposite the inter-plate spring space; spring/adjuster load therefore tends to seat the flange against the plywood rather than pull it away;
- drill the bushing bore 8.0 mm through the Baltic birch, preferably in a drill press with a sharp brad-point bit and sacrificial backing to minimize breakout;
- the bushing's 15.6 mm barrel (0.625 in overall including the flange) is fully contained within the 3/4 in plate, its rear end 3.4 mm below the rear face, and is bonded over its full cylindrical length;
- use WEST SYSTEM G/flex 650 epoxy for the wood-to-metal structural bond;
- bond directly to clean, fresh Baltic-birch bore surfaces; do not CA-harden or pre-seal the bushing bore;
- degrease and lightly abrade the outside of the bushing barrel before bonding without materially reducing its diameter;
- wet both the bore and bushing barrel with a thin continuous epoxy film, while keeping epoxy out of the internal 1/4 in-80 thread;
- seat the flange fully against the outboard face, remove squeeze-out from both faces, and cure with the bushing axis held square to the plate;
- the thin bushing flange is treated primarily as a seating/depth-stop feature. Structural retention is provided by the full-barrel epoxy bond and cylindrical bearing of the bushing in the plywood.

This arrangement adds no hardware thickness to the inter-plate gap.

10.3 Compression springs and seat washers

Components:

- McMaster 2022N186 compression springs
- McMaster 98017A199 stainless spring-seat washers

Each of the three springs is captured between two seat washers, producing:

- 3 springs
- 6 spring-seat washers
- 6 shallow spring-seat-washer counterbores

The counterbores register the washers; they are not spring pockets. Each washer is tacked into its counterbore with a small amount of G/Flex epoxy.

The nominal wood-face-to-wood-face inter-plate gap is 0.500 in / 12.7 mm at the neutral mirror position.

The spring-seat washer counterbores are:

- diameter: 5/8 in (Forstner cutter);
- nominal depth: 1/32 in / 0.03125 in / 0.794 mm;
- use the same nominal depth for all three seats in the mirror plate and all three corresponding seats in the stationary plate.

The McMaster 98017A199 washers are approximately 0.028–0.036 in thick, so the 1/32 in recess places them approximately flush with the plywood surfaces across their thickness tolerance. With the 0.750 in-free-length McMaster 2022N186 springs and the 0.500 in nominal inter-plate gap, the springs operate at approximately 0.250 in nominal compression before small washer-thickness and adjustment-position variations.

10.4 Spherical washer interfaces

Each of the three adjuster locations uses:

- McMaster 91131A028 spherical washer set
- one shallow close-fitting plywood socket/counterbore.

Total:

- 3 spherical-washer counterbores

These sockets participate in the mirror-cell gravity load path. The washer OD bears against the counterbore wall, transferring gravity load from the 3/4 in mirror plate/adjuster assembly to the 1/2 in stationary plate.

Target depth:

- approximately 1/16 in / 1.6 mm

Each socket is cut with a 1/2 in Forstner bit and lightly dressed with a file to a close free fit on the washer OD, not a loose retention pocket.

Stationary-plate adjuster through-hole

At each adjuster station, use a 1/8 in / 3.175 mm initial alignment guide hole through the 1/2 in stationary plate. This guide hole establishes a common center for the spring-seat counterbore on the mirror-facing face, the spherical-washer socket on the rear face, and the final adjuster clearance hole.

After machining both counterbores, enlarge the guide hole to a final 5/16 in / 0.3125 in / 7.94 mm through-hole. This is intentionally oversized relative to the 1/4 in Kozak TS250-80-2500 adjuster so the plywood does not become the angular bearing as the mirror plate tilts. A 5/16 in hole provides 0.03125 in radial clearance around a 0.250 in adjuster. Across a conservative full 0.500 in plate thickness, a 3° adjuster-axis tilt produces approximately 0.0262 in lateral displacement (0.500 × tan 3°), so the clearance accommodates approximately the full ±3° angular capability of the McMaster 91131A028 spherical-washer interface. The actual effective constraining thickness is somewhat less because of the shallow counterbores.

The machining sequence is: 1/8 in alignment guide hole → shallow counterbores from both faces → 5/16 in final through-hole. The spherical washer, rather than the plywood through-hole, remains the intended angular bearing and load-transfer interface.

CA conditioning

After cutting and fitting each spherical-washer socket:

1. wick a small amount of McMaster 1818A45 / Loctite 4061 low-viscosity CA into the exposed Baltic-birch fibers;
2. allow it to cure fully;
3. if the hardened socket is tight, lightly dress it back to the free fit.

The CA hardens the wood; it is not used to glue the washer in place.

Spherical-interface lubrication

Apply a very thin film of light grease only to the mating concave/convex spherical bearing surfaces of each McMaster 91131A028 washer set. Work the pair through its angular range and wipe away excess. This reduces friction and stick-slip as the mirror plate tilts during fine adjustment.

Keep the outside diameter and lower/load-transfer surfaces dry so the lower washer half continues to bear directly against the close-fit CA-hardened plywood counterbore. Keep grease away from plywood, adhesive-bonding areas, and RTV bonding surfaces. The grease is Dynatex NLGI Grade 2 white lithium grease.

10.5 Mirror mounting

The mirror is compliantly supported in a plywood mirror plate by six radial silicone pads, which are cast and cured in place between the mirror edge and the inner surface of the mirror plate aperture.

The 203 mm mirror is carried within the 3/4 in mirror-plate aperture. Its back surface is flush with the rear face of the plate, since both rest on the cure surface (§10.6) during bonding; its front surface is therefore recessed about 1 mm below the front face of the plate.

Six DOWSIL 737 RTV pads are placed radially between:

- the plywood aperture inner surface; and
- the mirror's outer glass edge.

These are compliant radial support/bond pads. Nominal pad geometry is approximately 18 mm circumferential length × 7 mm axial depth × the full approximately 1.7 mm radial bond-line thickness. Place the six pads at 60° intervals. Maintain the approximately 1.7 mm radial gap during cure with temporary non-stick centering shims positioned between pads; do not squeeze the mirror against the plywood or use the RTV pads themselves to force centering. The pad dimensions are practical build targets rather than precision-machined features. No counterbores or pockets are specified for the RTV pads.

Provisional: test adhesion to properly prepared Baltic birch before bonding, to decide whether an adhesion primer is necessary.

10.6 RTV glue-up cure surface

For mirror-plate glue-up and bench-top cure, use a non-stick polyethylene-family release surface beneath the assembly.

- approximately 12 × 12 in UHMW sheet
- large enough for the mirror plate (about 279 × 261 mm)
- flat, clean, dry
- no release spray required

This protects the workbench from accidental RTV creep or squeeze-out.

The release sheet is a shop supply rather than part of the finished instrument.

10.7 Angle brackets

Hardware joining the 1/2 in stationary plate to the 1/2 in base plate:

- 2 × McMaster 8681N11 3 in-long aluminum angle brackets (3 in legs, 1/4 in thick, Ø8.33 mm holes);
- 8 × McMaster 98164A527, 5/16 in-18 × 1-3/8 in 316 stainless button-head hex-drive screws;
- 8 × McMaster 90099A030 5/16 in-18 18-8 stainless heavy-profile nylon-insert locknuts;
- 8 × McMaster 96659A134 5/16 in stainless washers.

Each bracket is flush with a side edge of the plates, with its base leg on the base top extending forward and its upright leg against the front face of the stationary plate. Each leg has two holes, centered across the leg, 0.75 in and 2.25 in from the outer corner. The screws are pushed through from the bracket side, with the button heads on the bracket's inner faces; the washers and locknuts are on the opposite faces of the plywood, the rear face of the stationary plate and the underside of the base. Each bracket therefore takes four screws, two through the plate and two through the base, eight in all. The plywood holes match the bracket holes.

The button heads (Ø0.547 in × 0.166 in) are low enough that the heads of the two holes nearest the inside corner clear each other by more than 6 mm, so all four holes are used. The washer and locknut stacks are on the far sides of the plywood and cannot meet.

The 1-3/8 in screw suits the 1/2 in plywood + 1/4 in bracket + washer + 7/16 in locknut stack of about 1.25 in: the thread projects approximately 0.12 in, about 2 threads, beyond the locknut, preserving full locknut engagement without excessive overhang under the base or behind the stationary plate.

The bracket attachments are through-bolted rather than relying on wood screws. The locknut stacks under the base lie near its side edges, well outside the quick-release screw pattern on the underside (§10.9).

10.8 Mirror safety and transport retention

Front safety retention is intentionally non-stressing:

- loose retainers/fingers prevent escape of the mirror;
- they apply no deliberate optical preload;
- any surface that could touch glass should be soft/non-marring; adhesive-backed F1 felt (McMaster 8334K15) is the facing.

Transport sandwich

For shipping or travel the cell is disassembled and the mirror plate, with its mirror and bonded bushings, is held between two covers. The adjusters and springs are removed, and the stationary plate, with its seat washers, stays with the base plate and brackets.

- Each cover is 1/8 in hardboard (masonite or similar) cut to the profile of the mirror plate, with a clearance hole at each of the three adjuster-bushing centers. There is one cover on the front and one on the rear. The foam bears across the whole plate face, so the cover is supported over its area and needs no particular stiffness.
- The mirror's back surface is flush with the rear face of the plate (§10.5), so the rear cover carries a full-profile sheet of thin closed-cell foam, which bears evenly on the glass and the plywood.
- The front glass is recessed about 1 mm below the plywood face, and the bushing flanges stand 0.254 mm proud of it. The front cover carries a full-profile foam gasket with a 206.4 mm opening (the mirror-plate aperture, so the glass floating in its radial gap is never overlapped) and a clearance hole at each bushing center. The gasket bears on the plywood only and never touches the glass, so the recess does not affect its preload and nothing contacts the reflective surface. It also seals the cavity over the mirror as a dust gasket.
- Adhesive-backed F1 felt covers the inside face of the front cover, and the foam gasket is stuck to the felt. The felt is a backstop: its surface stands off the glass by the compressed foam thickness plus the 1 mm recess, about 3 mm, so a mirror that breaks free of its RTV pads travels only that far before meeting a soft surface and not bare hardboard. The standoff must stay at least about 2 mm so that the felt does not touch the glass while the sandwich is merely clamped. The felt limits travel but absorbs little energy; the foam and the outer packaging do that. Its surface hangs above the coating, so lift loose fibers from it with tape before assembly.
- Closed-cell foam is preferred, because open-cell foam sheds particles. Both gaskets are a uniform thickness of about 3 mm, compressed about 25–30% against the plywood; the compression sets the preload and the felt standoff.
- Three through-bolts, one at each bushing center, clamp the sandwich with a nut and washer on each cover. The bolts do not engage the bushing's 1/4 in-80 thread. Each passes through the bushing bore inside a snug polyethylene-family sheath that runs the length of the bushing, so the bolt never touches the fine thread. The 1/4 in-80 internal thread has a basic minor diameter of about 6.0 mm (calculated from the thread form, not measured), which bolt plus sheath must clear; an M3 or M4 bolt with a sheath of about 0.8–1 mm wall is the intended size. A fender-type washer under each nut spreads the load over the hardboard. The cover holes are just oversize to the bolt, so the bolts locate the covers on the plate.
- The covers touch the cell only through foam. The bolts tie the two covers to each other and do not couple either cover to the mirror plate, so a cover impact reaches the cell only after the foam. The only metal a cover could reach is the bushing's front flange, 0.254 mm proud of the plate face, and the foam must be crushed by about 90% before it does; the bushing's rear end is 3.4 mm below the rear face and cannot be reached. The foam is not stopped by a deliberate hard limit, because one would conduct shock.
- The bolts are all one length, about 38–40 mm: two covers, the felt, two compressed foam layers, the plate, and the nut stack. Each nut adjusts its own cover's foam compression independently. Thumb nuts make the assembly tool-free.
- The covers restrain the mirror axially only; lateral restraint is by the RTV pads. The rear foam holds the glass against rearward motion; forward motion is resisted by the RTV pads alone, with the front felt as the backstop. The sandwich is then wrapped in bubble wrap or boxed.

Provisional: the safety retainers are not yet designed, including the felt facing. For the transport sandwich: the foam material and thickness (about 3 mm uniform, holding the felt standoff at 2 mm or more); and the through-bolt size and length, the sheath material and wall, and the nuts, including the felt's thickness in the clamp stack. Measure a bushing bore before choosing the bolt and sheath.

10.9 Tripod interface

The mirror cell mounts on the tripod through a CAMVATE Manfrotto-type quick-release assembly with sliding plate (Amazon B0B2WBC921) on a Sunwayfoto LB-68R leveling base, whose top mounting stud is 3/8 in-16. The receiver stays on the LB-68R; the removable 577/501/504-compatible sliding plate stays attached to the underside of the 1/2 in Baltic-birch mirror-cell base.

The 130 g sliding plate is 7.00 in / 177.8 mm long, mounted flush with the front and rear edges of the 7.0 in-deep base, so it spans 0–7.00 in / 0–177.8 mm aft of the front edge and its geometric center lies 3.50 in / 88.9 mm aft of that edge. This is 0.5 in / 12.7 mm aft of the estimated mirror-cell center of gravity (§10.1), a small offset that the quick-release system's fore/aft sliding travel trims during final balancing.

The plate is attached by four McMaster 92125A196 M4 × 0.7 × 18 mm 90° flat-head screws:

flush flat-head M4 screw → countersunk 1/2 in Baltic-birch base → M4 × 0.7 tapped hole in solid aluminum of the sliding plate.

The screws are in a two-front / two-rear rectangular pattern. Each screw center is 1.875 in / 47.625 mm in from its nearest longitudinal plate end, and the two screws at each longitudinal station are ±0.500 in / ±12.7 mm from the plate centerline. The longitudinal stations are therefore 1.875 in and 5.125 in / 47.625 mm and 130.175 mm aft of the base front edge, with 1.000 in / 25.4 mm transverse spacing within each pair and 3.250 in / 82.55 mm between pairs. These locations lie in solid aluminum, clear of the sliding/clamping bearing surfaces and the central camera-screw slot. The heads finish flush with or slightly below the plywood top surface. The holes are tapped with an M4 × 0.7 tap (3.3 mm tap drill), and the screws are retained with Loctite 243 medium-strength removable threadlocker.

The sliding plate is used without its camera screws. Its camera face, with the slot, goes against the underside of the base. In the base, each screw has a Ø4.5 mm through hole, countersunk 90° to Ø8.4 mm at the top surface, which seats the Ø8 mm head about 0.2 mm below the surface. The 18 mm screw then reaches about 5.5 mm into the plate, roughly eight threads. The base is drilled from the layout below; the plate holes are then marked through the base holes with the plate held against the underside, and tapped.

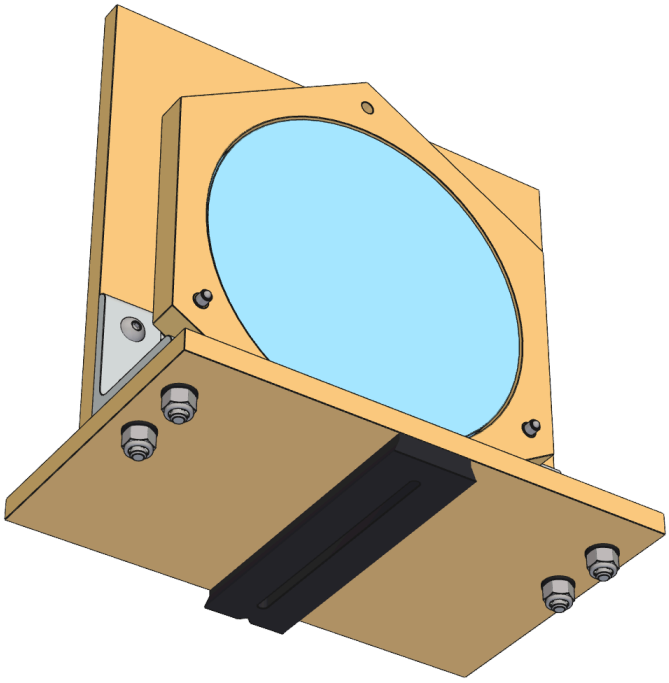


Fig. 17. Mirror cell from below: the quick-release sliding plate on the underside of the base, and the bracket locknuts

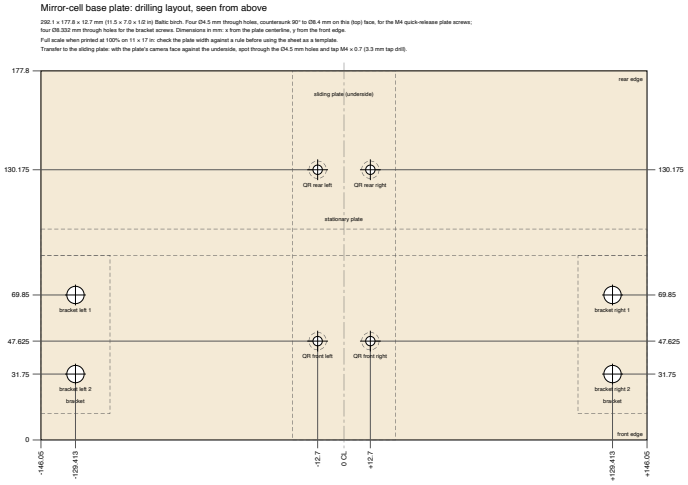


Fig. 18. Drilling layout of the base plate, seen from above, in cell coordinates (front edge at the bottom)

11. Provisional areas and next steps

Procurement state and the CAD state of printed parts are tracked in `bom/bom.csv`.

11.1 Provisional areas

Design areas marked **Provisional** in the sections above:

- **Light source** (§6.2, §6.3): heatsink temperature rise at 700 mA; white-module long-gap focus margin.
- **Lens cradle and phone rest** (§9): the lens alignment procedure, untested (§9.9). The alignment tolerances (§9.3) rest on an assumed lens position behind the cutoff plane and an assumed camera pupil; the rear-glass vignetting limit rests on an assumed distance from the rear glass to the camera pupil, to be fixed by a flat-field photograph.
- **Rail pivot bushings** (§4.3): the 8 mm bored pivot holes with press-in bronze sleeves, and the same in the pivot lugs' front holes; the bolt's thread length against the stack; the drilling template and frame model still show 5.5 mm pivot holes.
- **Slit stage head** (§7.6): the stage's dimensions are the manufacturer's, not measured; the blade overhang and the toe bar's squeeze are first-print values, as is how flat the stamped blade folds lie on the seats and under the toe pad (§7.1, §7.4); the stage-screw length behind the plate.
- **Baffle** (§7.8): the clamp's bore clearance and split gap; the net's stiffness and vane depth; the glare reduction of the painted vanes, to be confirmed on the bench.
- **Field lens** (§7.9): the iris-to-slit distance its position and the iris setting rest on (calculated, not measured); the mirror illumination with the lens in place and removed, to be confirmed from a flat-field photograph.
- **Slit focus target, ground-glass variant** (§7.7): the pocket clearance, gasket, and grit.
- **Cassette razor-blade clamping** (§8.7): how the clamp bars bear on a blade whose 7.04 mm fold lies under them, and the resulting bar height.
- **Mirror cell** (§10.5, §10.8): RTV adhesion primer decision; safety retainers; transport sandwich foam, through-bolts, sheaths, and nuts.

11.2 Next steps

1. Set focus on the slit blades with each LED module and confirm that the white module reaches focus; if not, add a thin cap-flange shim (§6.3). Bench-test the heatsink temperature rise.
2. With the field lens in place and removed, photograph the flat-field mirror illumination, confirm the iris setting, and measure the iris-to-slit distance (§7.9).
3. Bond the mirror into the mirror plate (§10.5) after an adhesion test.
4. Test the lens alignment procedure (§9.9), then write a consolidated setup and alignment procedure for the apparatus.
5. Design the mirror-cell safety retainers and settle the transport sandwich (§10.8).
6. Print the baffle clamp as a short coupon to settle its bore clearance and split gap, then print the baffle, paint it, wrap it, and compare the stray light with the baffle in place and removed (§7.8).